Basic Electrical and Electronics Engineering
Course Code: CSE1002

School of Computer Science , Engineering and Applications

Prepared By



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Course Name	CSE1002
Course Code	Basic Electrical and Electronics Engineering
Credits	4
Prerequisites	Basic knowledge of Physics, Elementary Mathematics
L-T-P-C	3-0-2-4

COURSE OBJECTIVES
1. To understand the fundamental concepts of electrical quantities, circuit elements, sources, and basic electrical circuits.
2. Explain the working principles and applications of semiconductor devices, diodes, rectifiers, and Zener diode voltage regulators.
3. Apply basic electrical and electronic concepts to interface sensors and Arduino
4. Analyze the operation of BJT and MOSFET as electronic switches
5. To implement basic operational amplifier circuits and understand their applications in amplification

COURSE SYLLABUS

•**Module I: Fundamentals of Electrical Circuits**

Introduction to electrical engineering, Basic electrical quantities: voltage, current, power and energy, Circuit elements: resistor, inductor and capacitor, independent voltage and current sources, Series and parallel circuits, Ohm's Law, Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL).

•**Module II: Semiconductor Devices and Applications**

Introduction to semiconductors: conductors, insulators and semiconductors, P-type and N-type semiconductors, PN junction diode: construction, working principle and V-I characteristics, Diode as a switch, Rectifiers: Half-wave and Full-wave/Bridge – working and applications, Zener diode as a voltage regulator.

•**Module III- Sensors and Arduino Applications**

Introduction to sensors and transducers, classification of sensors: analog and digital sensors, Applications of sensors in everyday life and computer engineering, Introduction to Arduino, Sensor Interfacing with Arduino: Interfacing LED and push button with Arduino, Interfacing LDR/light sensor, interfacing temperature sensor, Interfacing soil moisture sensor .

•**Module IV: Bipolar Junction Transistors (BJT) and MOSFET**

Introduction to BJT: construction, terminals and working principle, BJT characteristics, BJT as an electronic switch, BJT as an amplifier, Common Emitter, Common Base and Common Collector configurations, Introduction to MOSFET, N-channel and P-channel MOSFET – basic operation, MOSFET as an electronic switch, Comparison of BJT and MOSFET.

•**Module V: Operational Amplifiers and Applications**

Introduction to operational amplifier, differential amplifier and ideal Op-Amp characteristics, inverting and non-inverting amplifier configurations, voltage follower, adder and subtractor, Op-Amp as comparator, OP-AMP as integrator and differentiator.

TEXTBOOKS

1. “Electronic-devices”-by-floyd-9th-edition.Prentice Hall
2. "Principles of Electrical Engineering and Electronics" by V.K. MEHTA, ROHIT MEHTA Publisher S Chand

REFERENCES

1. David A. Bell, “Electronic Devices and Circuits”, Oxford University Press, 5th Edition, 2008.
2. D.P. Kothari, I. J. Nagrath, “Basic Electronics”, McGraw Hill Education (India) Private Limited, 2014.

LABORATORY	
Sr. No.	Content
1	Identification and Measurement of Basic Electronic Components
2	Verification of Kirchhoff's Laws (KVL & KCL) – Apply KVL in a series circuit and KCL in a parallel circuit, and verify through measurements.
3	Diode as a Switch and Zener Diode as Voltage Regulator
4	Study of Half-Wave Rectifier – Observe input and output waveforms, and calculate ripple factor.
5	Study of Full-Wave Rectifier (Center-tap & Bridge) – Compare performance of full-wave rectifiers with half-wave rectifier.
6	Introduction to Arduino UNO – Familiarization with Arduino UNO, Arduino IDE, digital/analog pins and basic programming; interface an LED and push button.
7	LDR-Based Light Sensing using Arduino – Interface an LDR with Arduino and control an LED based on light intensity.
8	Basic Op-Amp Applications – Implement and verify inverting, non-inverting amplifier and voltage follower circuits.
9	BJT as a Switch – Demonstrate switching action of a transistor using LED or small load.
10	Mini project

EVALUATION SCHEME

Theory: 60%			
Continuous Assessment	Mid-term	End-term	Total
25	25	50	100

Practical: 40%				
Continuous Assessment			End-term Practical	Total
Lab Manual	Lab Assessment	Internal Viva-voce		
10	10	30	50	100

COURSE OUTCOMES At the end of the course the student will be able to:	
CO1	Explain basic electrical quantities, components, Ohm's Law, KCL and KVL.
CO2	Explain the working and characteristics of semiconductor devices
CO3	Apply Arduino to interface and experiment with basic sensors
CO4	Design and analyze rectifier and regulator circuits for low-level power supply applications.
CO5	Implement and evaluate operational amplifier circuits

ELECTRICITY IN OUR LIFE



Mobile Phone
Communication



Laptop
Work & Study



Air Conditioner
Comfort



Fan
Cooling



Light Bulb
Illumination



Television
Entertainment





Mobile Robot
Automation & Research



Refrigerator
Cooling & Storage



Washing Machine
Cleaning



Mixer / Grinder
Food Preparation



Electric Vehicle
Transportation



Solar Power
Renewable Energy



Computer
Information Processing

BASIC ELECTRICAL TERMINOLOGY

Voltage (V)



Potential difference between two points in a circuit.

Unit: Volt (V)

Current (I)



Flow of charge through a conductor per unit time.

Unit: Ampere (A)



Power (P)



Rate at which electrical energy is used or transferred.

Unit: Watt (W)

Energy (E)



Electrical work done or energy consumed.

Unit: Joule / kWh

All Together



These basic electrical quantities are the foundation of all electrical engineering concepts.

ELECTRIC CHARGE

Electric charge is the **fundamental property** responsible for electrical phenomena.

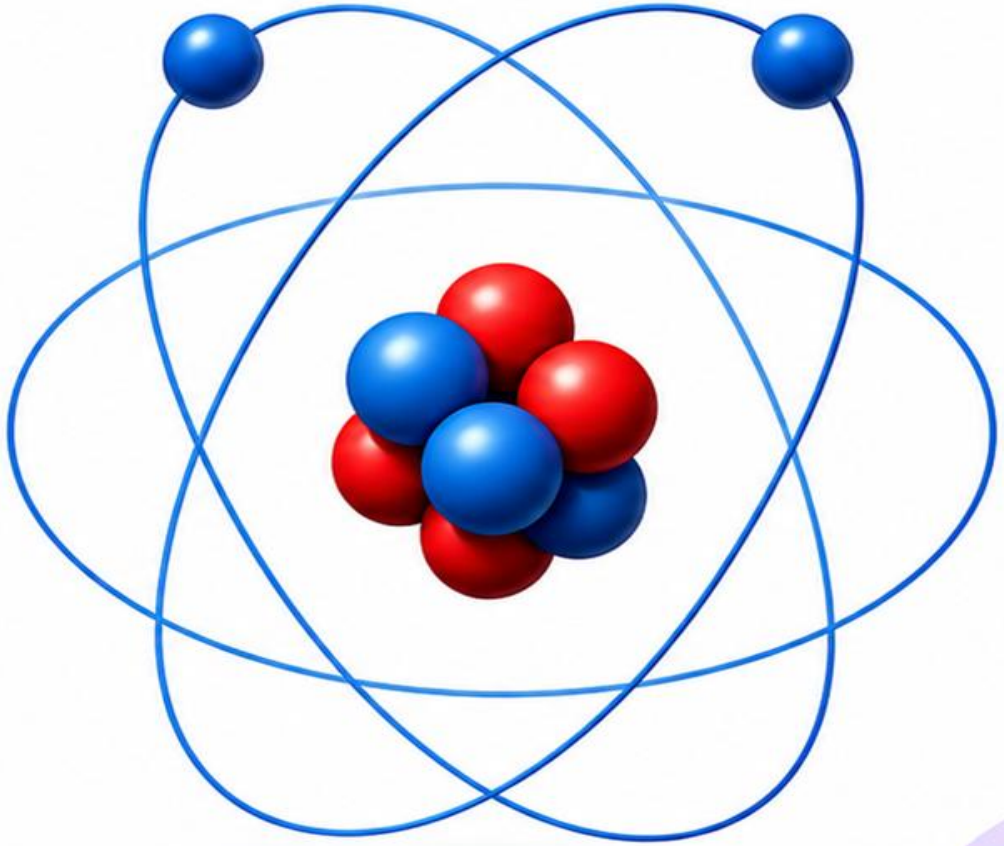
Symbol: Q

Unit: Coulomb (C)

Charge of an electron,

$$e = 1.602 \times 10^{-19}\ C$$

Positive and negative charge.

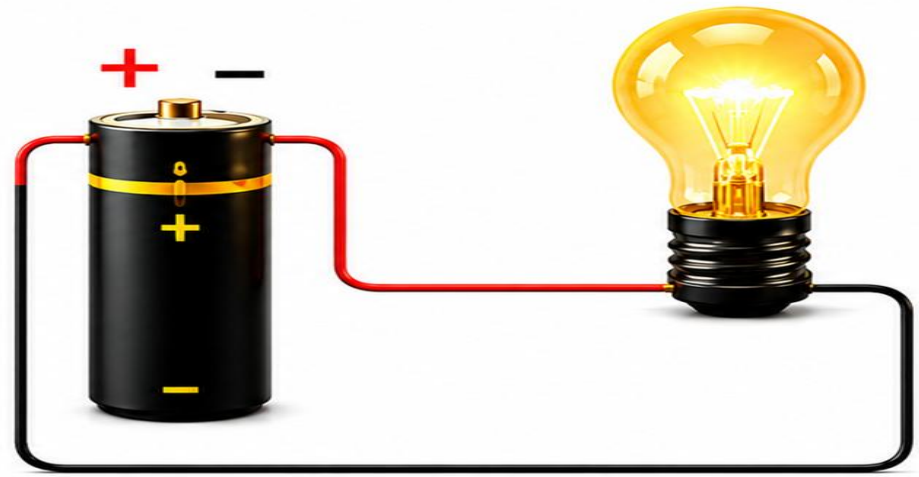


ELECTRIC CURRENT

Electric current is the **rate of flow** of **electric charge.**

$$I = \frac{Q}{t}$$

Where, I = Current (A)
Q = Charge (Coulomb)
t = Time (s)



Rate of flow of charge

- I = Q/t
- Unit: Ampere (A)
- Conventional current flows + to −.

VOLTAGE / POTENTIAL DIFFERENCE

Voltage is the **work done** per unit charge.

$$V = \frac{W}{Q}$$

Where,

V = Voltage (V)



Work done per unit charge

- V = W/Q
- Unit: Volt (V)
- Provides electrical potential difference.

Example:- A 5 V USB supply provides a 5-volt potential difference that allows current to flow through a connected device.

ELECTRICAL POWER

Electrical power is the rate at which electrical energy is transferred or consumed.

$$P = VI$$





- Rate of electrical energy transfer
- P = VI
- $P = I^2R$
- $P = V^2/R$
- Unit: Watt (W)

Example:- A 100 W bulb operating for 5 hours per day calculate energy/day?
Sol:- Energy/day = 100 × 5 = 500 Wh = 0.5 kWh
For 30 days = 15 kWh.

Energy is the ability of a physical system to perform work or cause change.

Kinetic

Energy of motion

Potential

Stored energy

Chemical

Energy in bonds

Electrical

Energy carried by electricity

Thermal

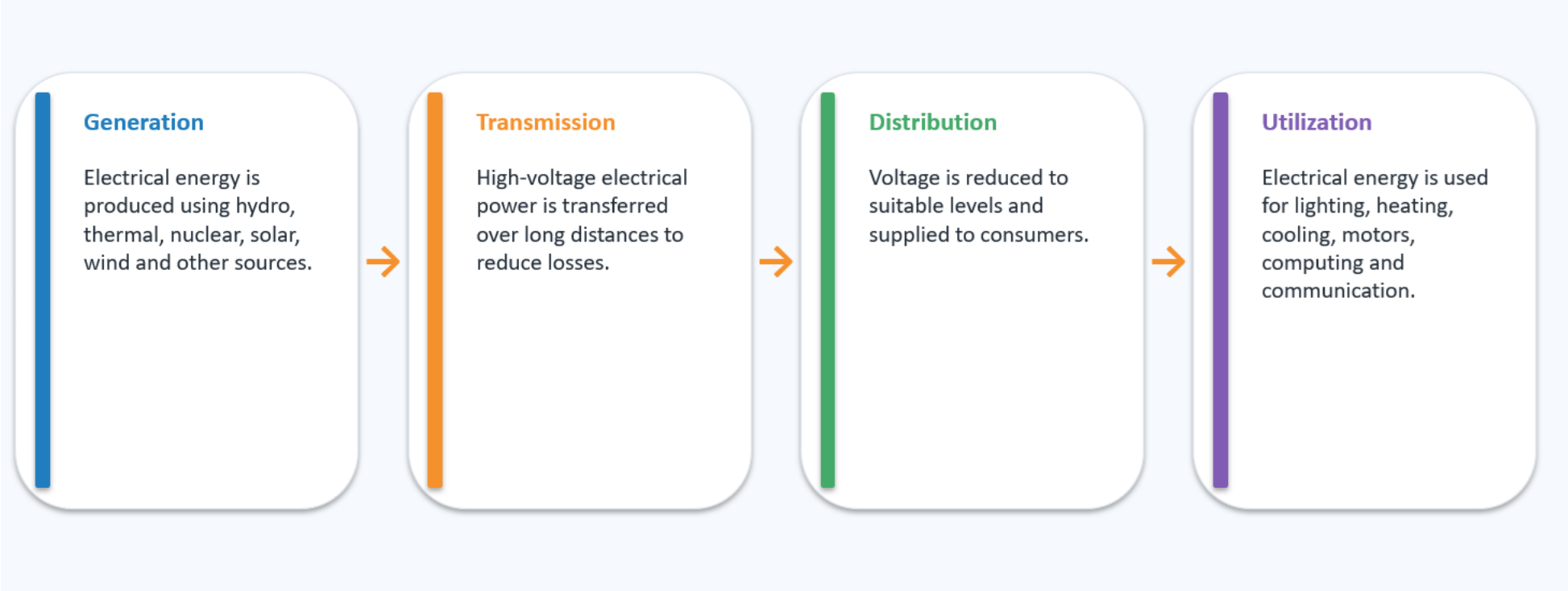
Related to temperature

Nuclear

Energy stored in the nucleus

SI unit of energy: Joule (J)

How Electrical Energy reaches to us



ELECTRICAL ENERGY

Electrical energy is the total amount of electrical work consumed.

$$E = Pt = VIt$$

Units : Joule (J)

E = Electrical energy (J)

P = Power (W)

t = Time (s)



- E = Pt = VIt
- Units: Joule, Wh, kWh
- 1 kWh = 3.6×10^6 J
- Used for electrical energy consumption.

Example:-

A 100 W Bulb operates for 5 hours per day.

Calculate :-

1. Energy consumed per day
2. Energy consumed 30 days.

Solution:-

100W lamp * 5h =500 Wh/day.

=0.5 kwh/day.

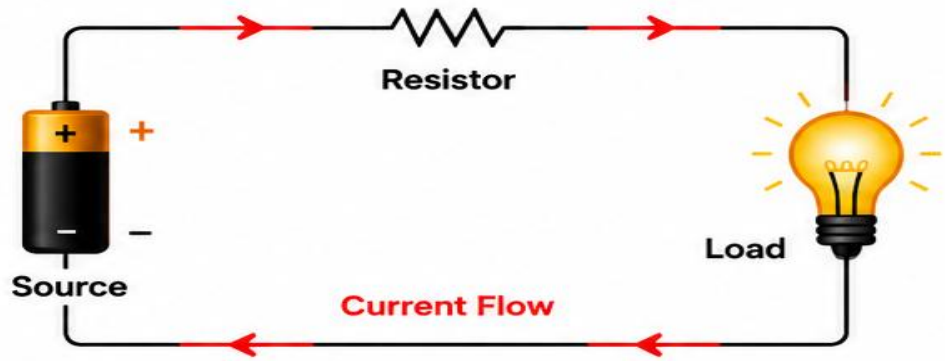
For 30 days = 15kwh.

https://www.tinkercad.com/things/ewjeSHZerAn-led-intensity-variation/editel?returnTo=%2Fthings%2FewjeSHZerAn-led-intensity-variation

ELECTRICAL CIRCUIT

WHAT IS AN ELECTRICAL CIRCUIT?

An electrical circuit is a closed path formed by connecting electrical components such that electric current can flow through it.



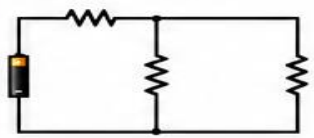
CLASSIFICATION OF ELECTRICAL CIRCUITS

1. BASED ON CONNECTION

- **Series Circuit**

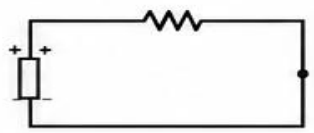


- **Parallel Circuit**



2. BASED ON NATURE OF CURRENT

- **DC Circuit**

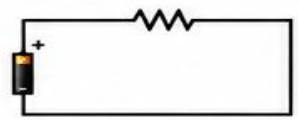


- **AC Circuit**

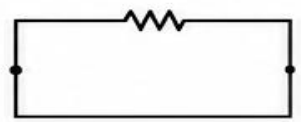


3. BASED ON ELEMENTS USED

- **Active Circuit**
  (Contains active elements like source)

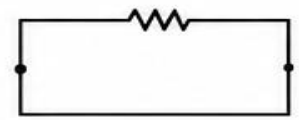


- **Passive Circuit**
  (No active source)

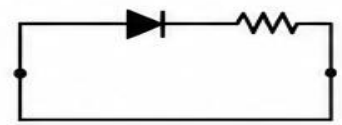


4. BASED ON TIME VARIATION

- **Linear Circuit**
  (Parameters are constant)

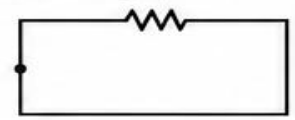


- **Non-linear Circuit**
  (Parameters vary)

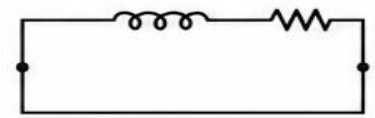


5. BASED ON PHYSICAL LAYOUT

- **Lumped Circuit**
  (Size is small compared to wavelength)

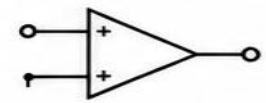


- **Distributed Circuit**
  (Size is comparable to wavelength)

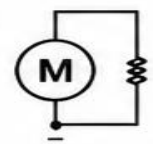


6. BASED ON APPLICATION

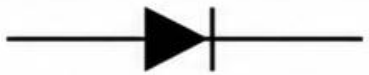
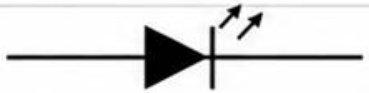
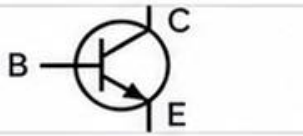
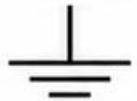
- **Signal Circuit**
  (For signal/data processing)



- **Power Circuit**
  (For power transmission and usage)



Physical Component vs Circuit Symbol Chart

Component	Physical Part	Circuit Symbol	Main Function
Resistor			Limits current; drops voltage
Capacitor (non-polarized)			Stores charge; blocks DC, passes AC
Capacitor (polarized/electrolytic)			Stores charge; blocks DC, passes AC (polarity sensitive)
Inductor			Stores energy in a magnetic field; opposes changes in current
Diode			Allows current in one direction only
LED			Emits light when forward biased
NPN Transistor			Amplifies or switches current
SPST Switch			Opens or closes a circuit
Battery			Provides DC voltage
Ground	Reference node / not a discrete physical component		Reference point (0 V) for a circuit

Active and Passive Elements

Definition:- Active elements supply or deliver energy to a circuit.

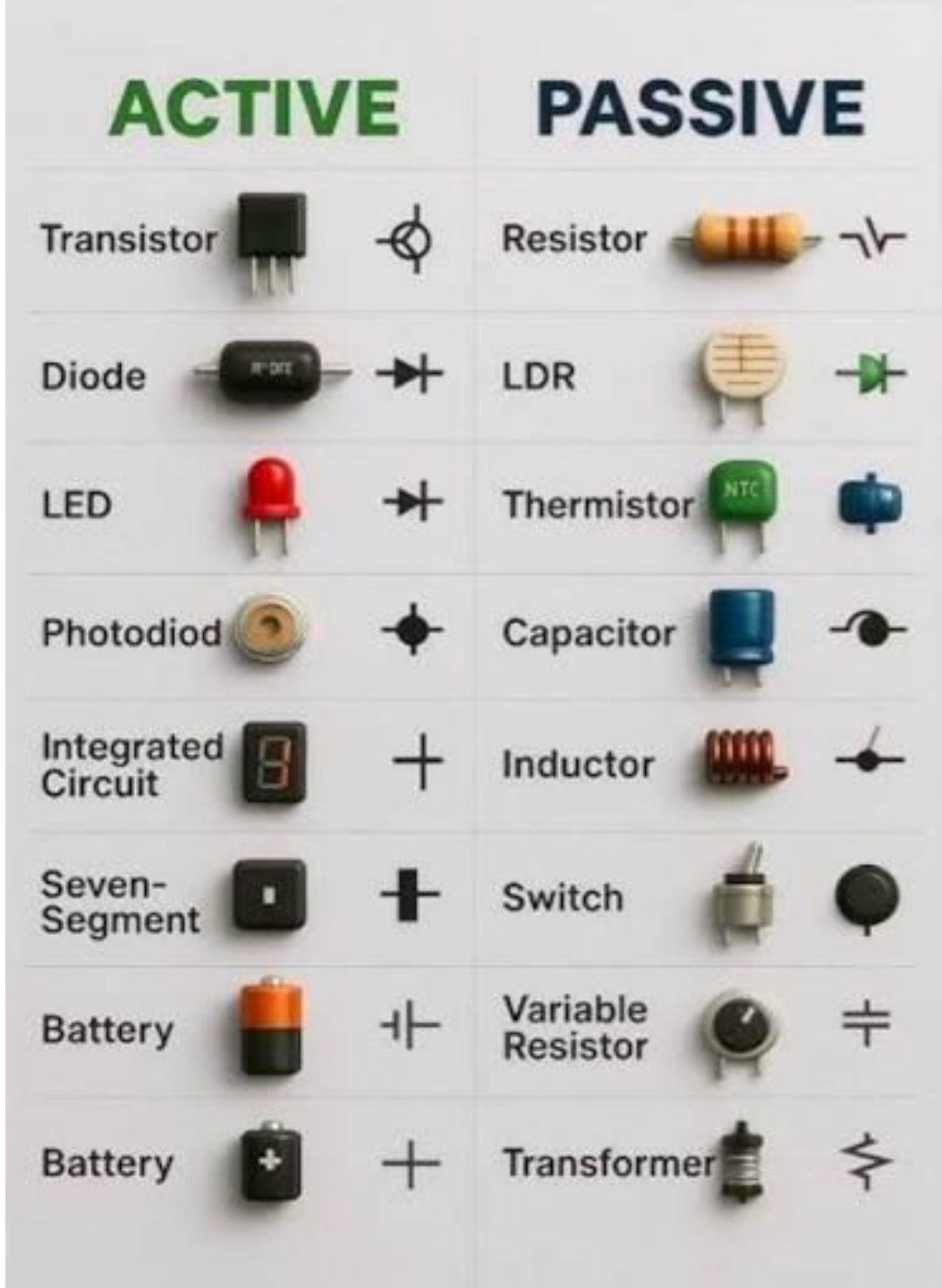
Active Elements :-

- **Energy Donors:** They supply electrical power or produce voltage and current.
- **Control Flow:** They can control or amplify the flow of electric current.
- **Examples:** Voltage sources, current sources, generators, transistors, and diodes.

Definition:- Passive elements absorb, store, or dissipate energy.

Passive Elements :-

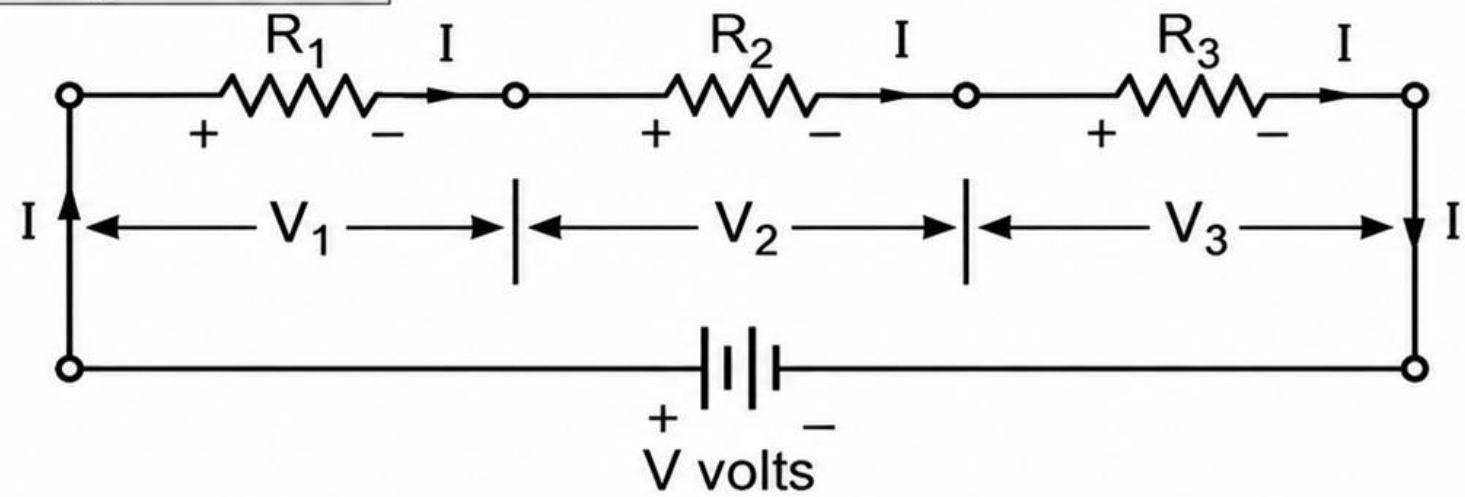
- **Energy Acceptors:** They receive power and cannot generate or amplify signals.
- **Storage and Dissipation:** They use energy, turn it into heat, or store it in electric and magnetic fields.
- **Examples:** Resistors, capacitors, and inductors.



Parameter	Active Elements	Passive Elements
Definition	Elements that can **supply energy** to a circuit or provide power gain.	Elements that **cannot generate energy**; they only absorb or store energy.
Energy	Can deliver energy to the circuit.	Absorb or store energy.
Power Gain	May provide power gain or amplification.	Cannot provide power gain.
Examples	Voltage source, current source, transistor, operational amplifier	Resistor, inductor, capacitor
Function	Supply, control, or amplify electrical signals.	Dissipate or store electrical energy.
Dependency	Sources can operate independently of the circuit response.	Their behavior depends on the applied voltage/current.
Applications	Amplifiers, power supplies, signal generation and processing	Filtering, energy storage, current limiting, coupling

SERIES CIRCUIT

Current same
voltage division



- The resistance R1, R2 and R3 are said to be in series

$$Req = R1 + R2 + R3$$

The total or equivalent resistance of series circuit is the arithmetic sum of resistance connected in series.

Characteristics of Series Circuits

1. The same current flows through each resistance.
2. The equivalent resistance is equal to the sum of the individual resistances
3. The voltage drop across each resistor will be different.
4. The supply voltage V is the sum of the individual voltage drops across the resistances.

$$V = V1+V2+\ldots\ldots\ldots +Vn$$

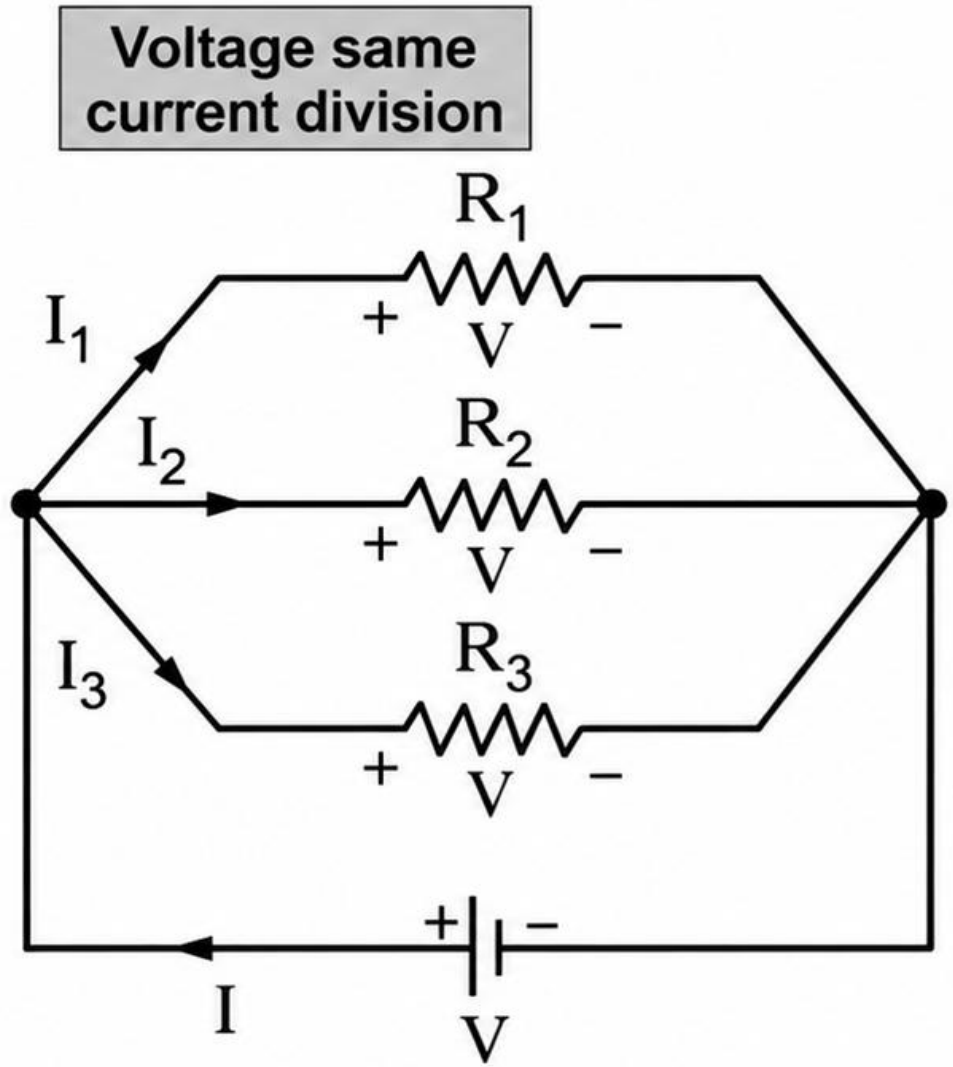
PARALLEL CIRCUIT

If Req is the equivalent resistance

$$\frac{1}{Req} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3}$$

If there are only 2 resistors

$$\mathbf{Req} = \frac{R1R2}{R1+R2}$$



Characteristics of Parallel Circuits

1. The potential difference across each resistor is the same.
2. The current through each resistor is different
3. The total current is always sum of all the branch currents.

$$I = I_1 + I_2 + I_3 + \ldots\ldots + I_n$$

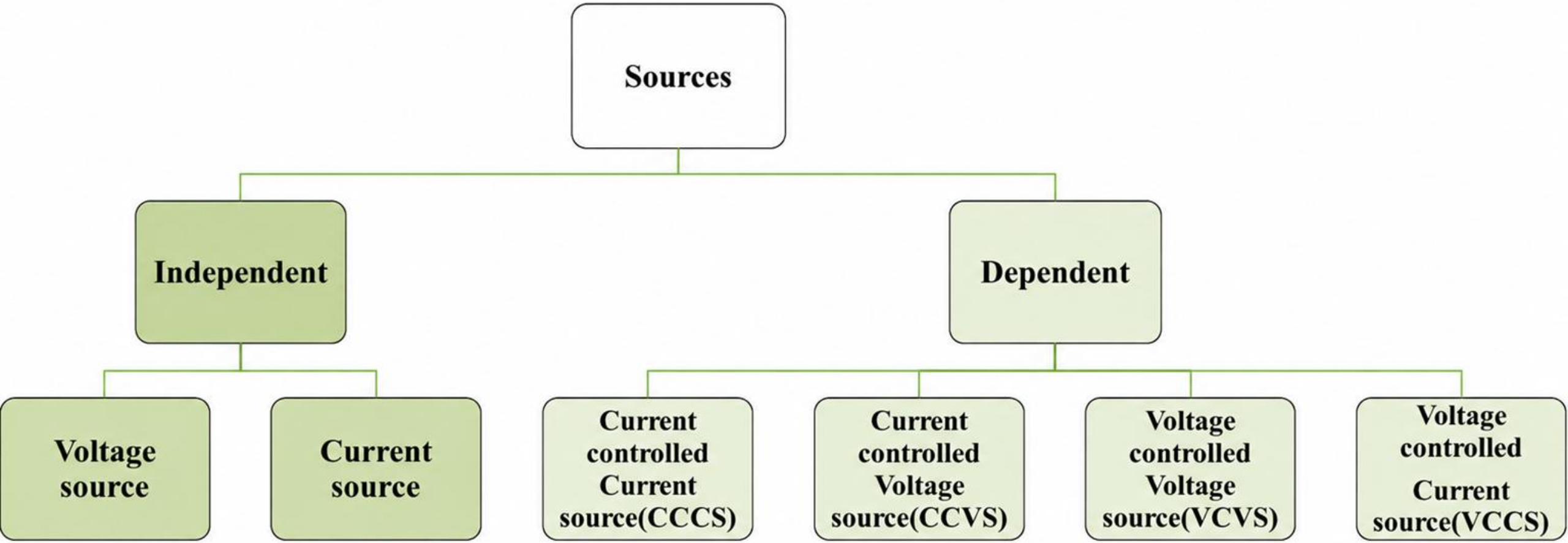
1. Resistance of the circuit is $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots\ldots + \frac{1}{Rn}$

Example:- House hold Appliances

- Voltage Across each branch is same.
- Current divides between branches.
- If one branch is open current can still flow throw other branch.

CLASSIFICATION OF SOURCES

A source is an electrical element that supplies energy to a circuit.



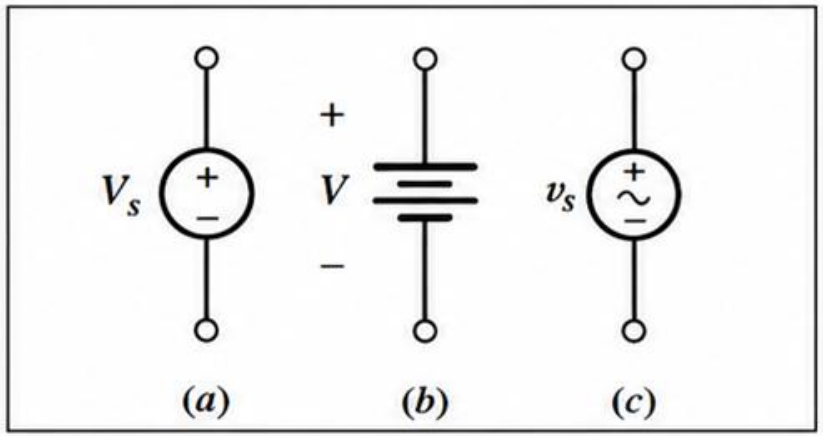
Battery

Type	Full Form	Controlling Quantity	Output
VCVS	Voltage Controlled Voltage Source	Voltage	Voltage
VCCS	Voltage Controlled Current Source	Voltage	Current
CCVS	Current Controlled Voltage Source	Current	Voltage
CCCS	Current Controlled Current Source	Current	Current

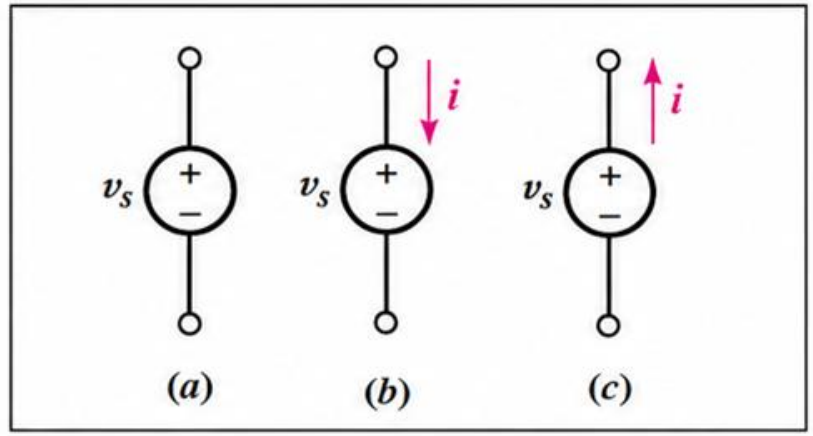
Independent Vs. Dependent Sources

Sources in which **the voltage is completely independent of the current, or the current is completely independent of the voltage**; these are termed **independent sources**.

Special kinds of sources for which either the **source voltage or current depends upon a current or voltage elsewhere in the circuit**; such sources are referred to as **dependent sources**.



(a) DC voltage source (b) battery symbol
(c) AC voltage source symbol



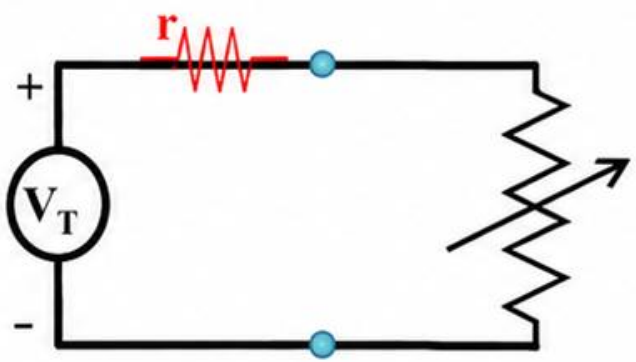
Circuit symbol of the independent voltage source

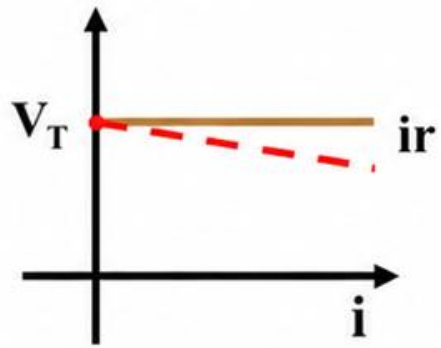
An independent voltage source is characterized by a terminal voltage which is completely independent of the current through it.

Independent Voltage Sources

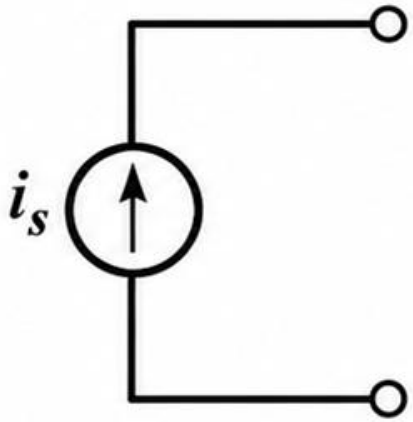
- ✓ The independent voltage source is an ideal source and does not represent exactly any real physical device.
- ✓ Because the ideal source could theoretically deliver an infinite amount of energy from its terminals (delivery of infinite current at a particular constant voltage).
- ✓ This idealized voltage source does, however, furnish a reasonable approximation to several practical voltage sources.

Ideal Vs. Practical Voltage Sources





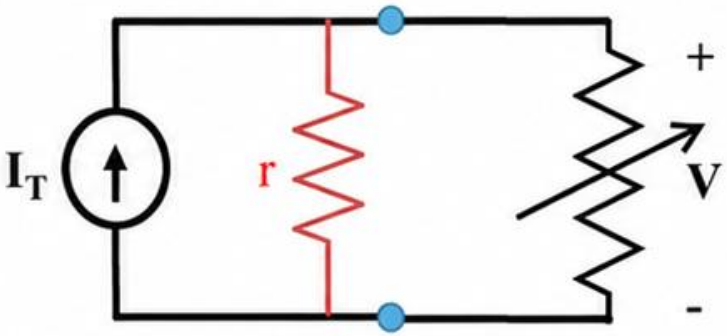
Independent Current Sources

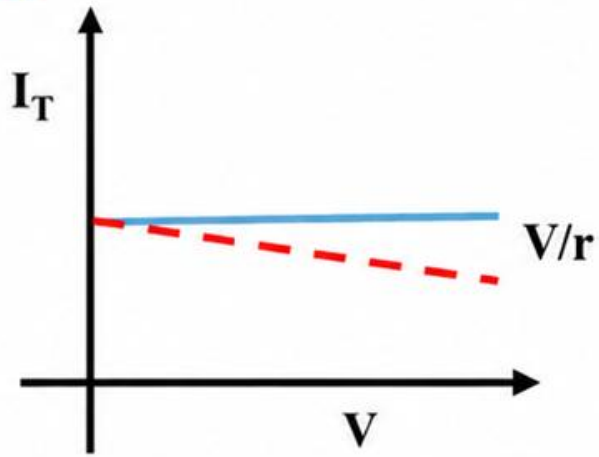


For independent current source, the current through the element is completely independent of the voltage across it.

Circuit symbol for the independent current source.

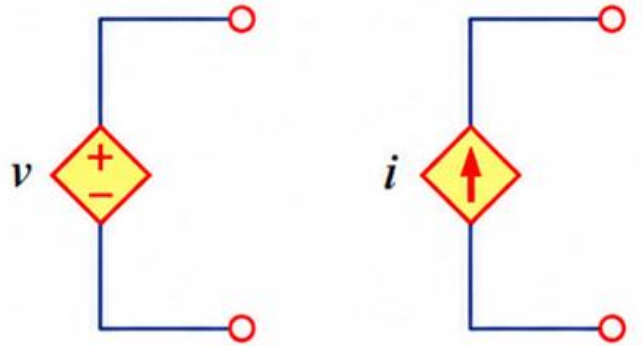
Ideal Vs. Practical Current Sources





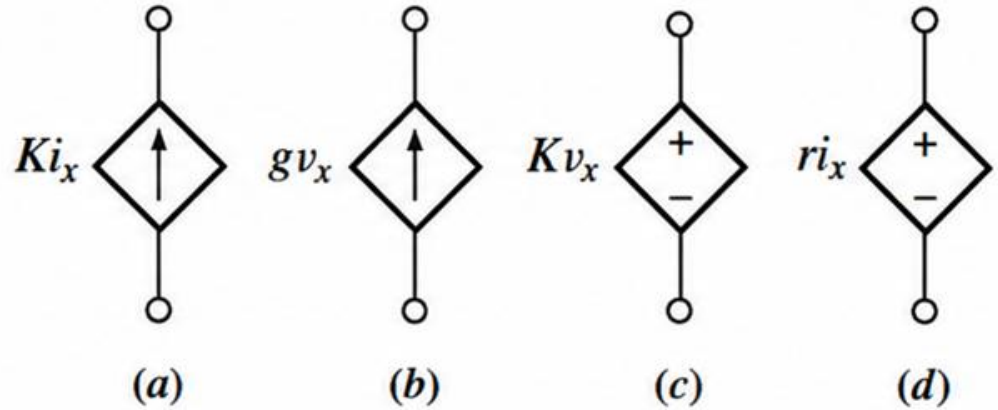
Dependent Sources

✓ **Dependent sources are useful in modelling elements such as transistors, operational amplifiers, and integrated circuits.**

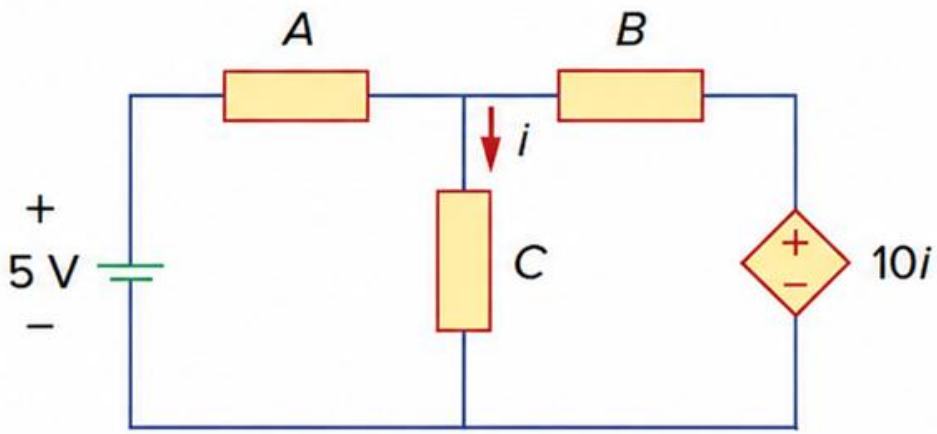


(a) dependent voltage source

(b) dependent current source



The four different types of dependent sources:
(a) current-controlled current source;
(b) voltage-controlled current source;
(c) voltage-controlled voltage source;
(d) current controlled voltage source.



The source on the right-hand side is a current-controlled voltage source.

A source produces 5 V regardless of what happens elsewhere in the circuit.

Answer: Independent Voltage Source

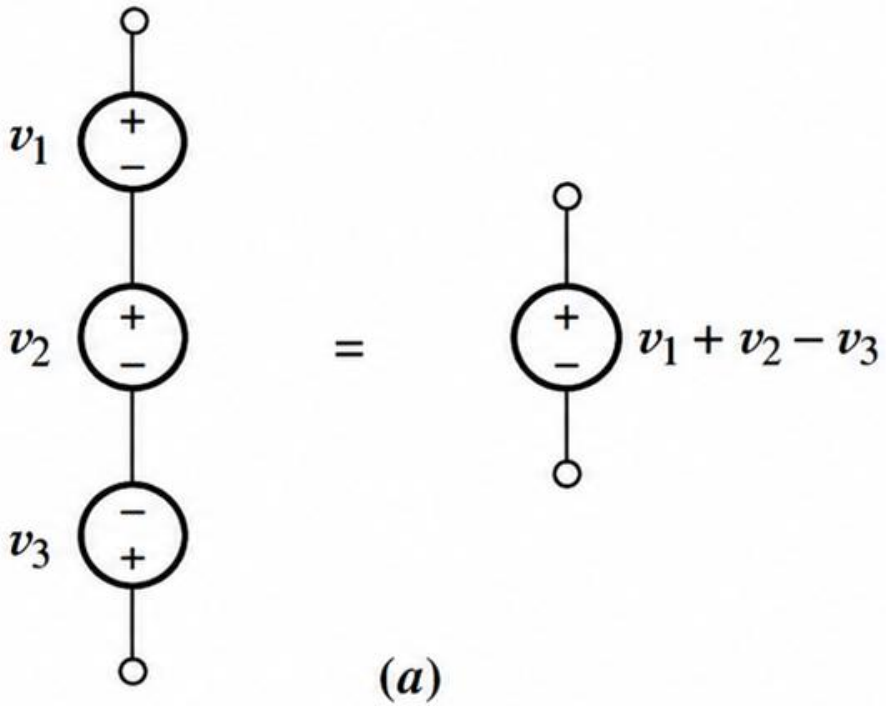
A source produces current equal to 2Vx

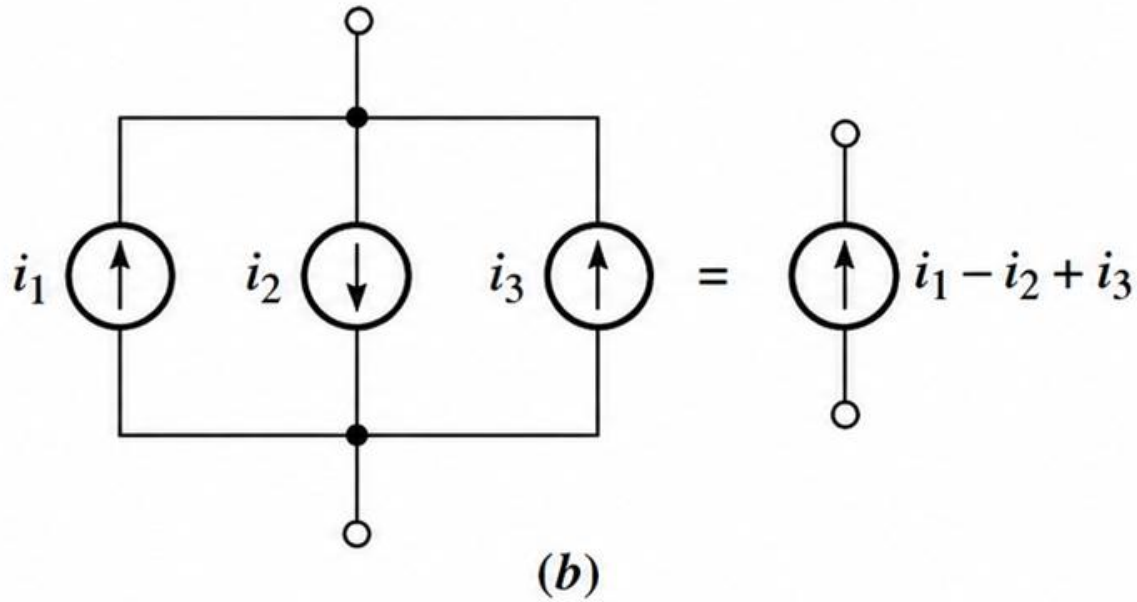
Answer: VCCS

A source produces voltage equal to 4Ix.

Answer: CCCS

SERIES AND PARALLEL CONNECTION OF SOURCES

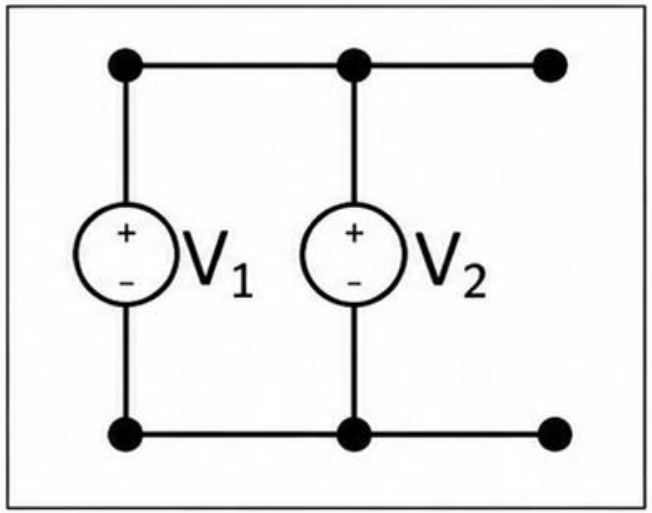




(a) Series-connected voltage sources can be replaced by a single source

(b) Parallel current sources can be replaced by a single source.

Voltage Sources in Parallel

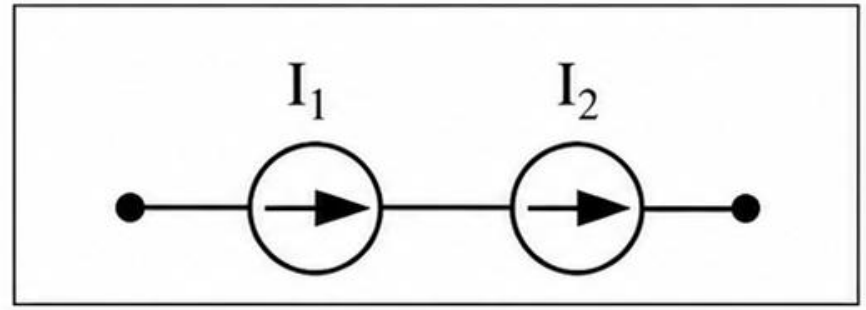


$V_1 = V_2$ ✓

$V_1 \neq V_2$ X

- Two voltage sources are connected in parallel to increase current rating.
- Two unequal voltage sources parallel connection should be avoided as it leads to circulating current among the sources

Current Sources in Series

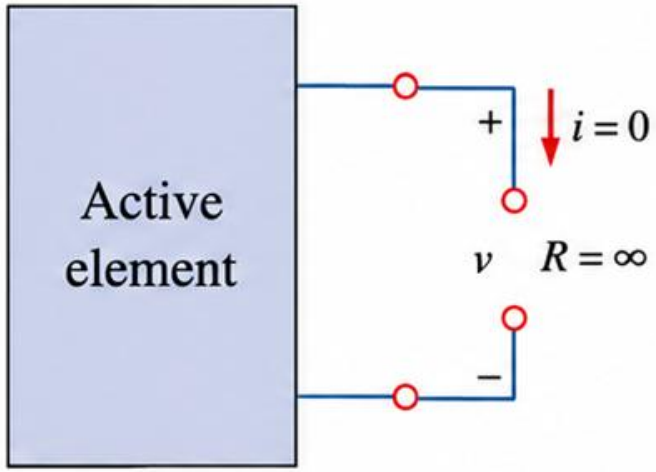


$I_1 = I_2$ ✓

$I_1 \neq I_2$ X

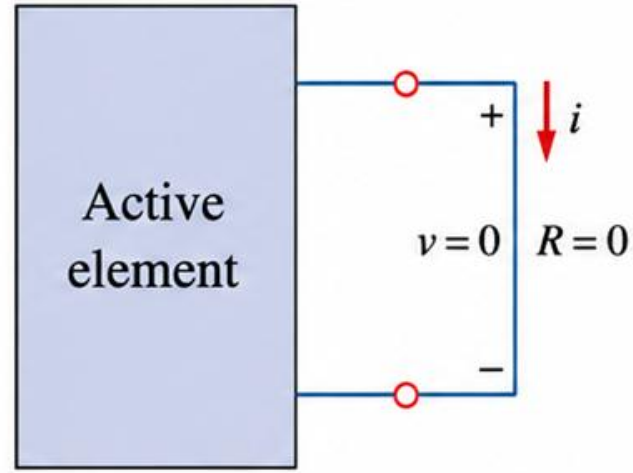
- Two different current sources must not be connected in series as it violates Kirchhoff's current law.

Termination



Open Circuit

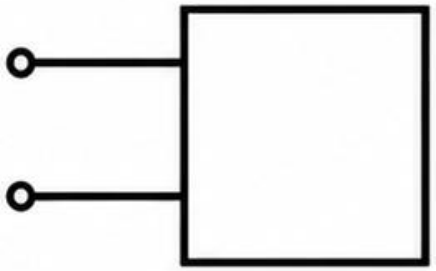
$R \to \infty$ $\quad$ $i = 0$



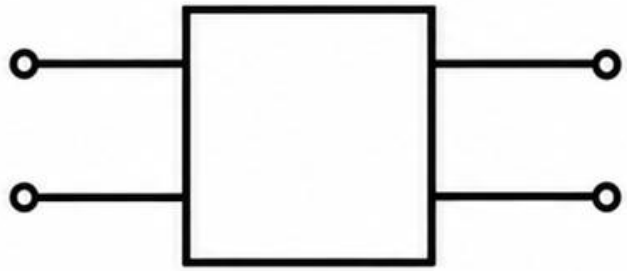
Short Circuit

$R \to 0$ $\quad$ $V = 0$

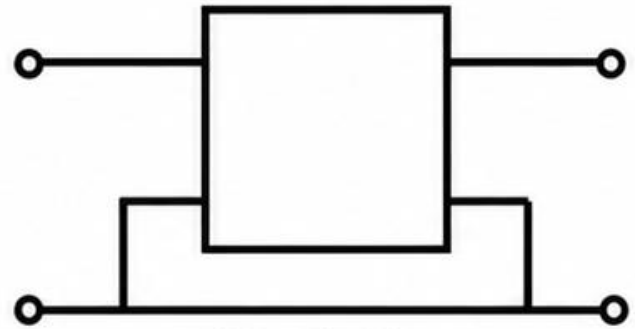
Concept of Terminals



2T ; 1- Port



4T ; 2- Port



3T ; 2- Port

Ohm’s Law

Statement:- Ohm’s law states that Current is directly proportional to the potential difference across a conductor, provided temperature and other physical conditions remain constant.

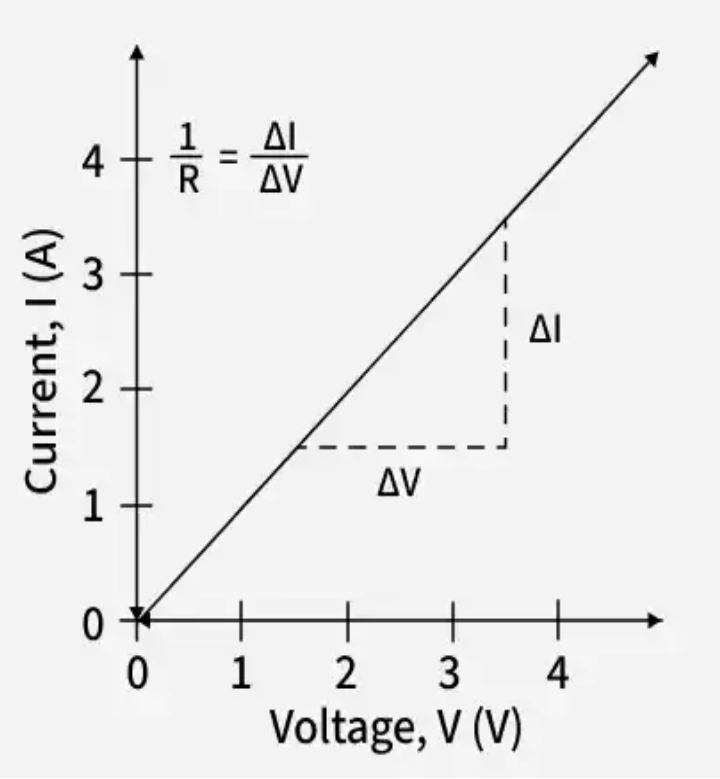
It describes the fundamental relationship between voltage and current in an electric circuit.

V∝I

V=I×RV=I×R

where,

- **R** is the Constant of Proportionality known as Resistance,
- **V** is the Voltage applied, and
- **I** is the current flowing through the electrical circuit.



Vector Form

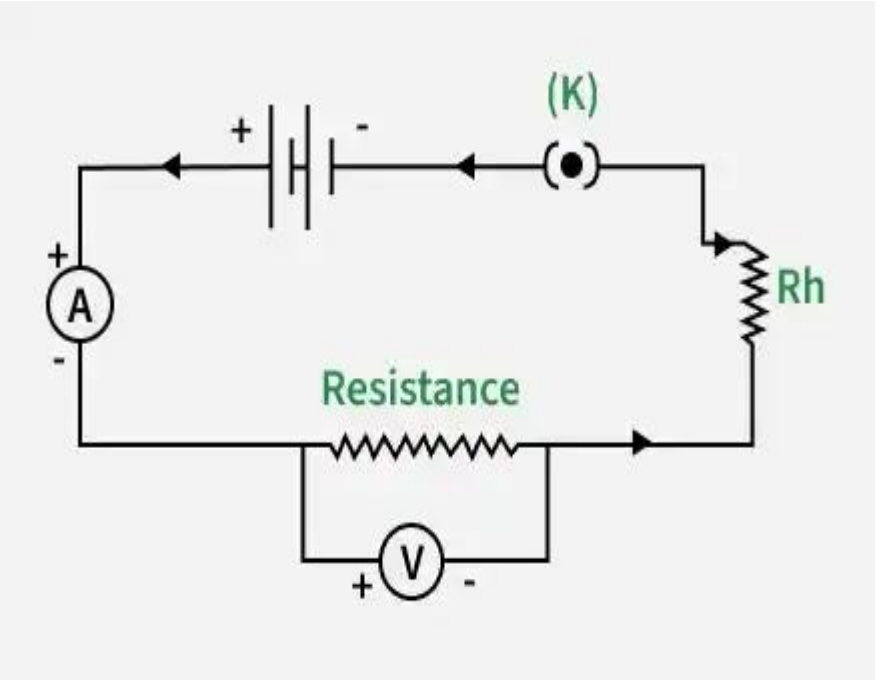
The relation between current and voltage is established by, Ohm's law, and its vector form is,

$$\vec{J}=\sigma\vec{E}$$

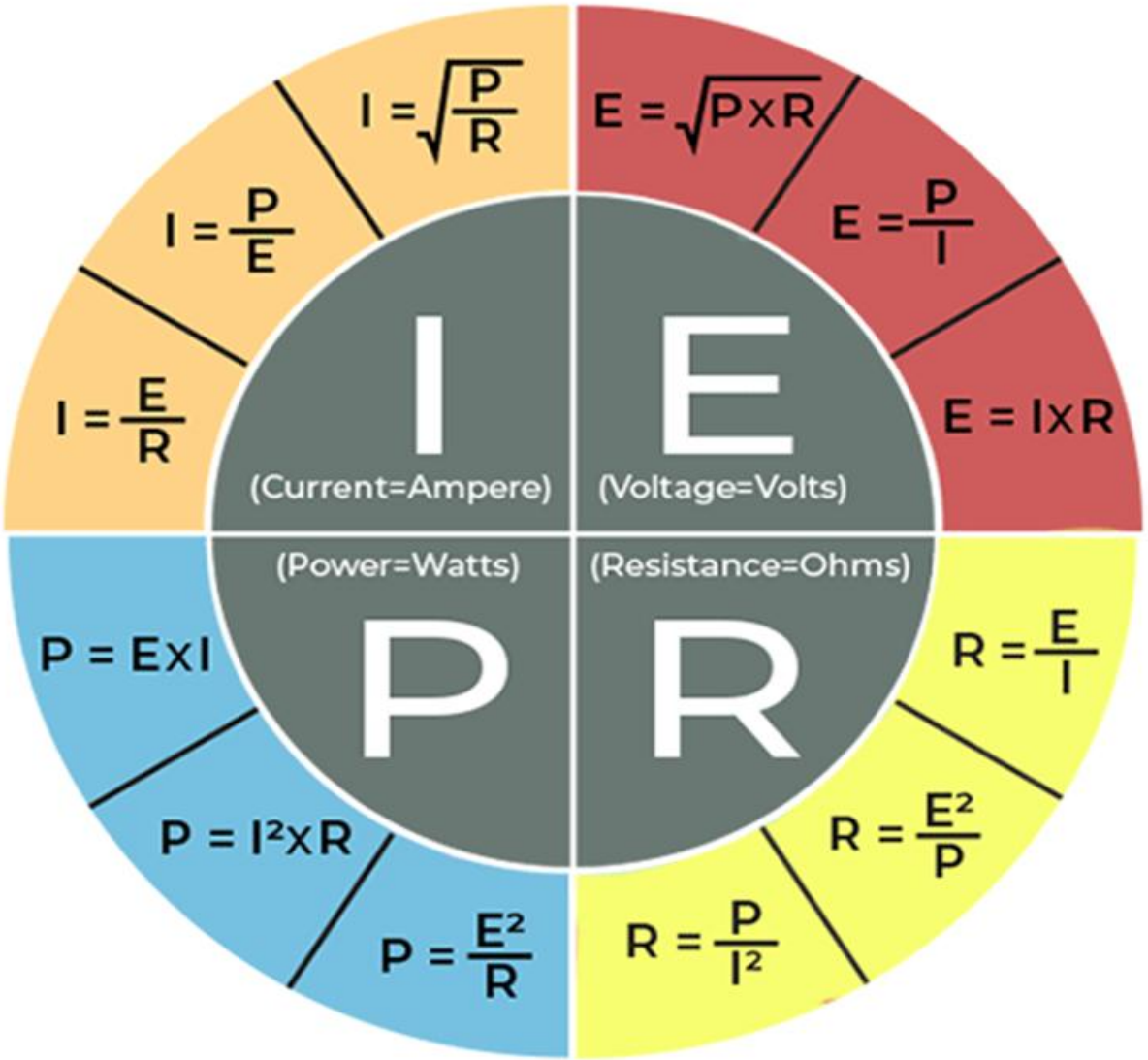
Where,

$\vec{J}$ is Current Density vector,
$\vec{E}$ is Electric Field vector, and
σ is conductivity of material.

Circuit Diagram



Pie Chart


I
(Current=Ampere)
E
(Voltage=Volts)
P
(Power=Watts)
R
(Resistance=Ohms)
$I = \sqrt{\frac{P}{R}}$
$I = \frac{P}{E}$
$I = \frac{E}{R}$
$E = \sqrt{P \times R}$
$E = \frac{P}{I}$
$E = I \times R$
$P = E \times I$
$P = I^2 \times R$
$P = \frac{E^2}{R}$
$R = \frac{E}{I}$
$R = \frac{E^2}{P}$
$R = \frac{P}{I^2}$

Applications of ohms law

1. **LED bulbs**– Helps determine the current required for an LED circuit and select a suitable resistor.
2. **Electric heater** – Used to calculate current drawn by the heating element.

3.**Electric fan** – Helps estimate current through the motor circuit under suitable operating conditions.

4. **Mobile chargers** – Used in designing and analyzing resistive parts of charging circuits.

4. **Battery-powered device** – Helps estimate the current drawn by a load from a battery.

4. **Automobile electrical systems** – Used for analyzing lamps, resistive loads, and wiring.

4. **Computers and electronic devices** – Used in basic circuit design and troubleshooting.

4. **Household appliances** – Helps calculate current consumption and select appropriate wiring and protective devices.

Limitations:

1. Not directly applicable to **diodes and LEDs**.
2. Not applicable to **transistors** in general.
3. Resistance may change with **temperature**.
4. Not suitable for many **non-linear devices**.
5. Simple Ohm’s Law is insufficient for **capacitors and inductors in AC circuits**.

Problems

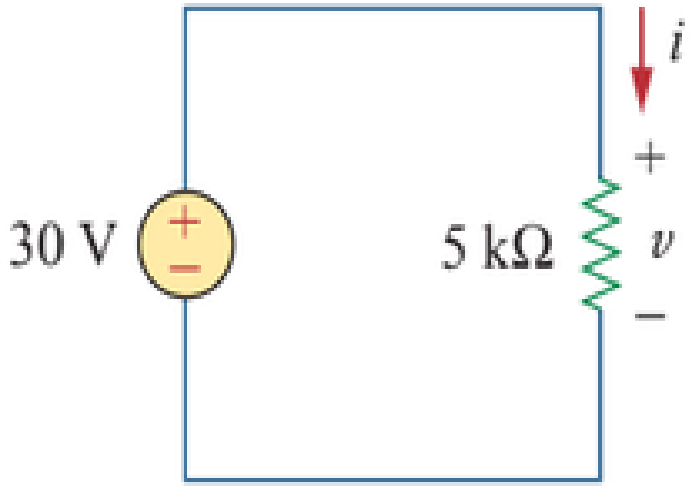
Question 1: Find the resistance of an electrical circuit with a voltage supply of 15 V and a current of 3 mA.

Question 2: If the resistance of an electric iron is 10 Ω and a current of 6 A flows through the resistance. Find the voltage between two points.

Question 3: Find the current passing through the conductor drawing 20 volts when the power drawn by it is 60 watts.

Question 4: A battery of 6 V is connected to the bulb of resistance 4 Ω. Find the current passing through the bulb and the circuit's power.

Question 5. In the circuit shown in Fig. calculate the current i, the conductance G, and the power p.



Solution:

The voltage across the resistor is the same as the source voltage (30 V because the resistor and the voltage source are connected to the same pair of terminals. Hence, the current is

i v/=R = 30/5 *10^3 = 6 mA

The conductance is

G = 1/R = 1/5 *10^3 = 0.2 mS

We can calculate the power in various ways using different equations:-

a] p = vi = 30(6 * 10^-3) = 180 mw.

b] p = i^2R = (6 * 10^-3) 5 * 10^3 = 180 mw.

c] p = v^2G = (30)2 *0.2 *10^-3 = 180 mW

INTRODUCTION

- Current gets divided in different path when resistances are connected in parallel, hence current division is applicable in parallel circuit.
- Voltage gets divided in series but current remain same, hence voltage division is applicable in series circuit.
- Currents and voltages follow KCL and KVL laws in circuits at junction points, nodes and in loops.
- Complex electrical circuits with multiple loops can be solved via mesh analysis

Current/Voltage Division finds applications in construction of electric meters and sensor devices





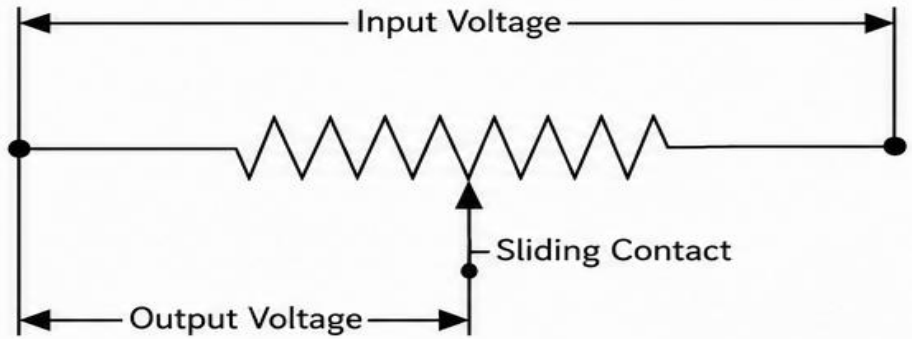




VOLTAGE DIVISION CONCEPT

The voltage division concept is used in various electronic circuits and devices to **obtain a desired output voltage** or **adjust signal levels**.

1. USED IN RHEOSTAT AND POTENTIOMETERS







2. USED IN RADIOS, AMPLIFIERS AND ELECTRONIC DEVICES

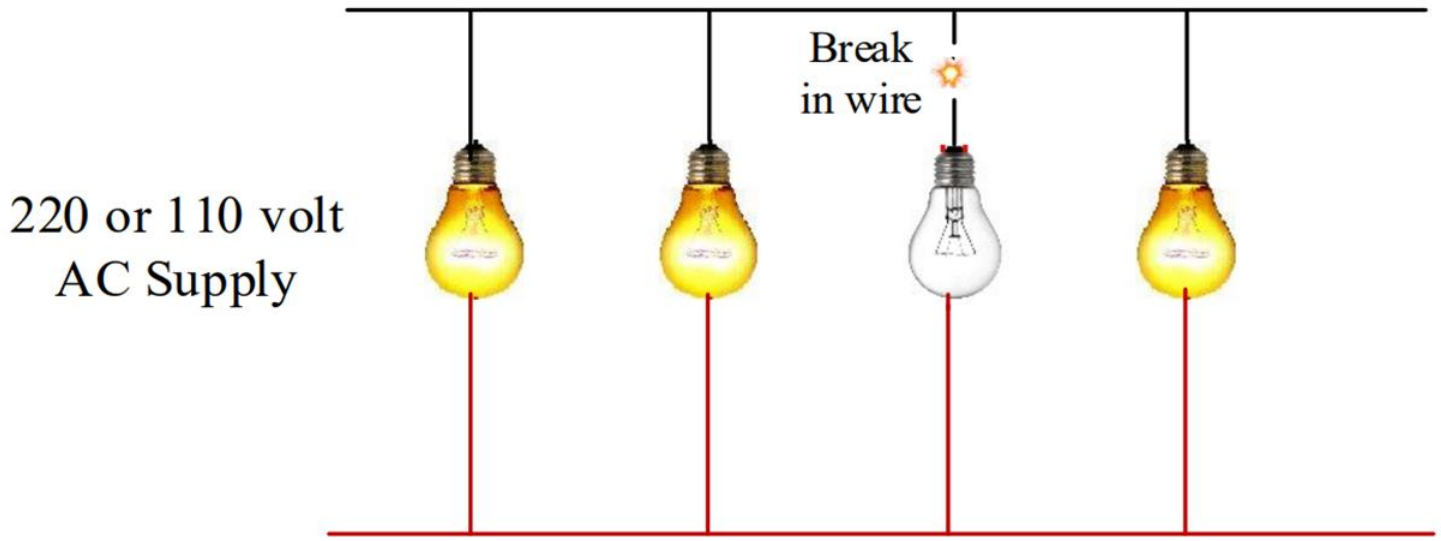
Voltage division concept is used in making **radios, amplifiers and electronic devices** for adjusting signal levels.



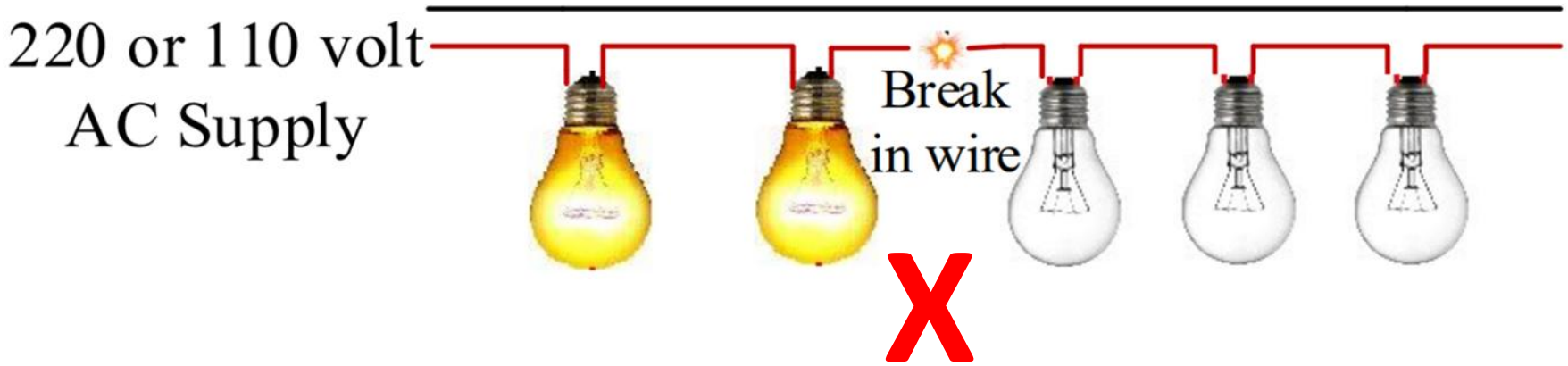




Lighting System



Rest of the bulbs are ON except the parallel path with break in circuit.



Rest of the bulbs are OFF after breaking point. Lighting System Series Connection Correct?

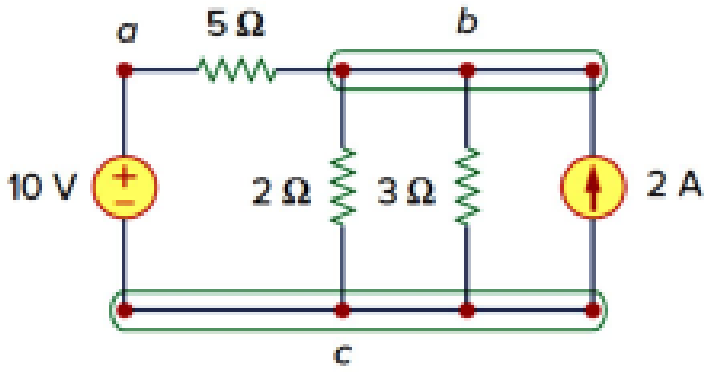
No! These 2 bulbs will also be OFF. There will be no current on the entire line because of open circuit.

NODE , BRANCHES AND LOOPS

A branch represents a single element such as a voltage source or a resistor in a circuit.

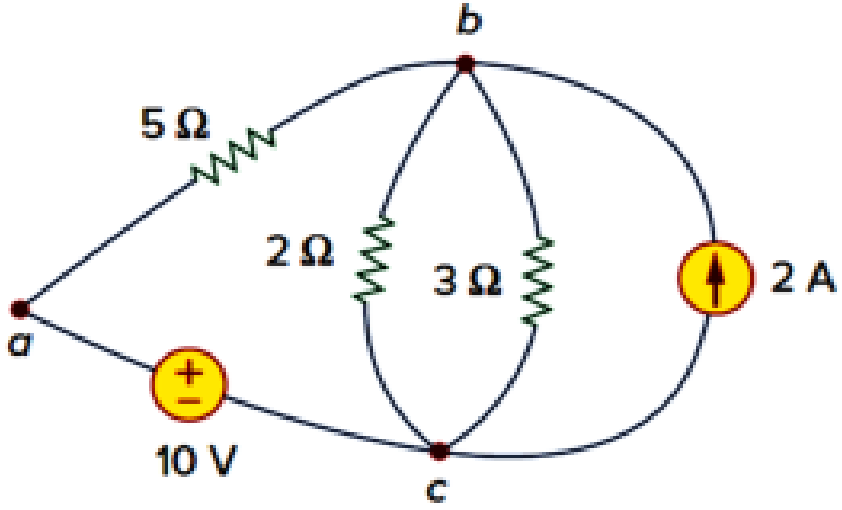
A node is the point of connection between two or more branches

A loop is any closed path in a circuit.



5 branches and 3 nodes

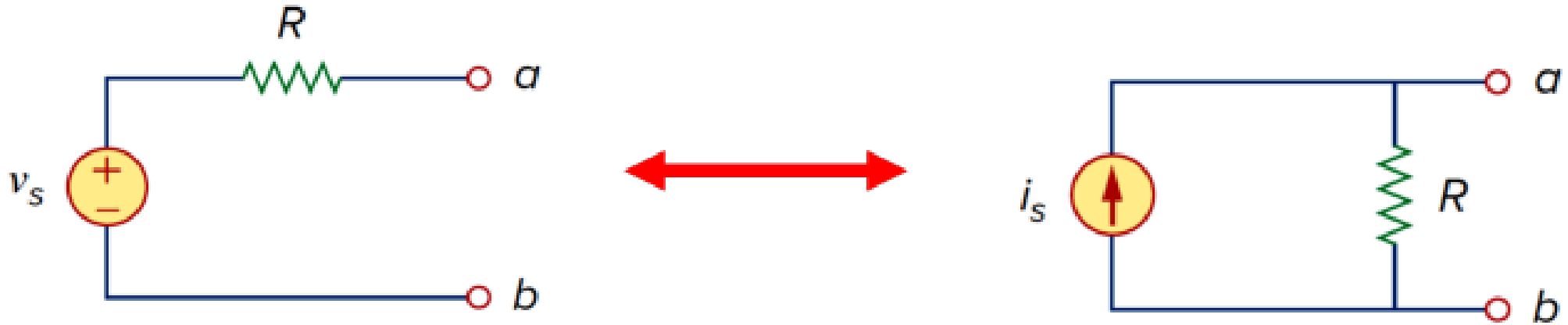
Same Circuit reconfigured



A network with *b branches, n nodes, and l independent loops* will satisfy the fundamental theorem of network topology:

$$b = n + l - 1$$

SOURCE TRANSFORMATION



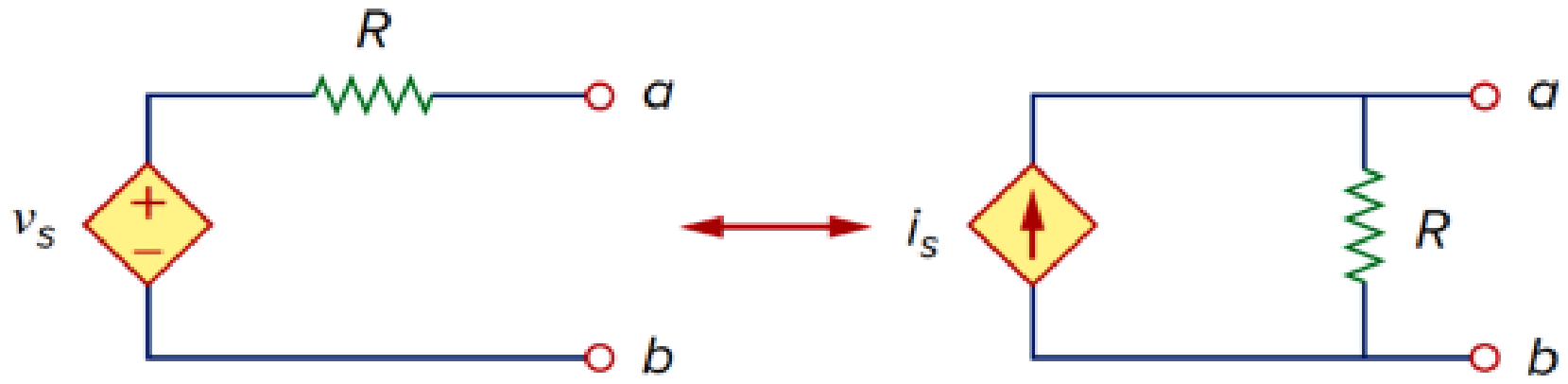
Transformation of independent sources.

$$v_s = i_s * R \text{ or } i_s = \frac{v_s}{R}$$

A *source transformation* is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R, or vice versa.

- **Source transformation also applies to dependent sources.**

- **A source transformation does not affect the remaining part of the circuit.**



One should keep the following points in mind when dealing with source transformation:

1. **The arrow of the current source is directed toward the positive terminal of the voltage source.**

2. **Source transformation is not possible when R = 0, which is the case with an ideal voltage source.**

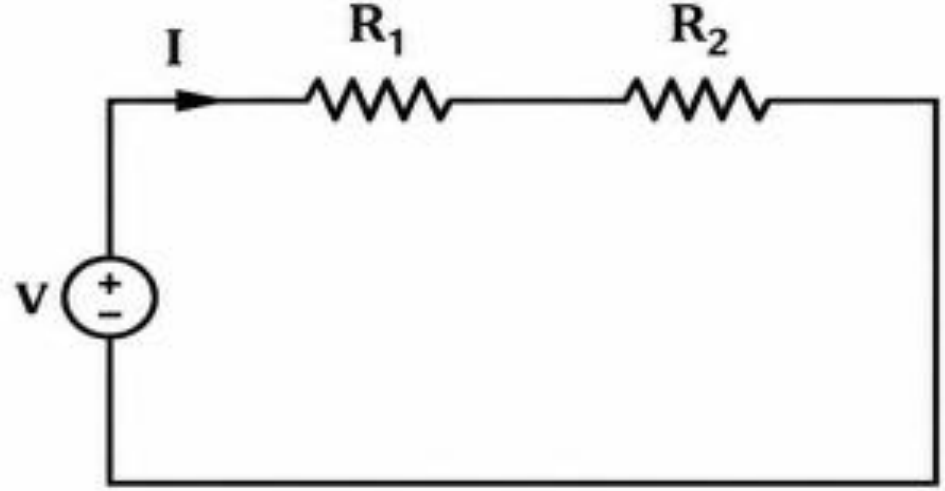
   However, for a practical, non-ideal voltage source, R ≠ 0.

3. **Similarly, an ideal current source with R = ∞ cannot be replaced by a finite voltage source.**

VOLTAGE DIVISION IN SERIES CIRCUIT OF RESISTORS

$$V_{R1} = \frac{V}{R_1 + R_2} \cdot R_1 = \left[\frac{R_1}{R_1 + R_2}\right] V$$

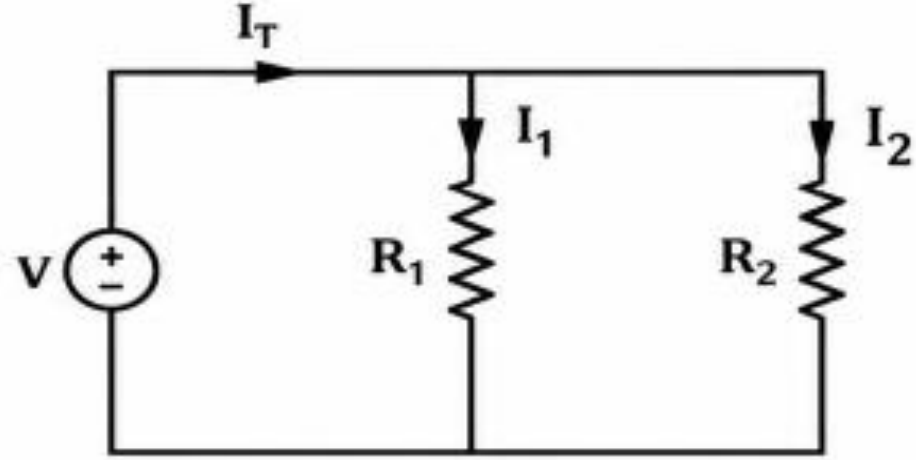
$$V_{R2} = \frac{V}{R_1 + R_2} \cdot R_2 = \left[\frac{R_2}{R_1 + R_2}\right] V$$



CURRENT DIVISION IN PARALLEL CIRCUIT OF RESISTORS

$$I_1 = I_T \left(\frac{R_2}{R_1 + R_2}\right)$$

$$I_2 = I_T \left(\frac{R_1}{R_1 + R_2}\right)$$

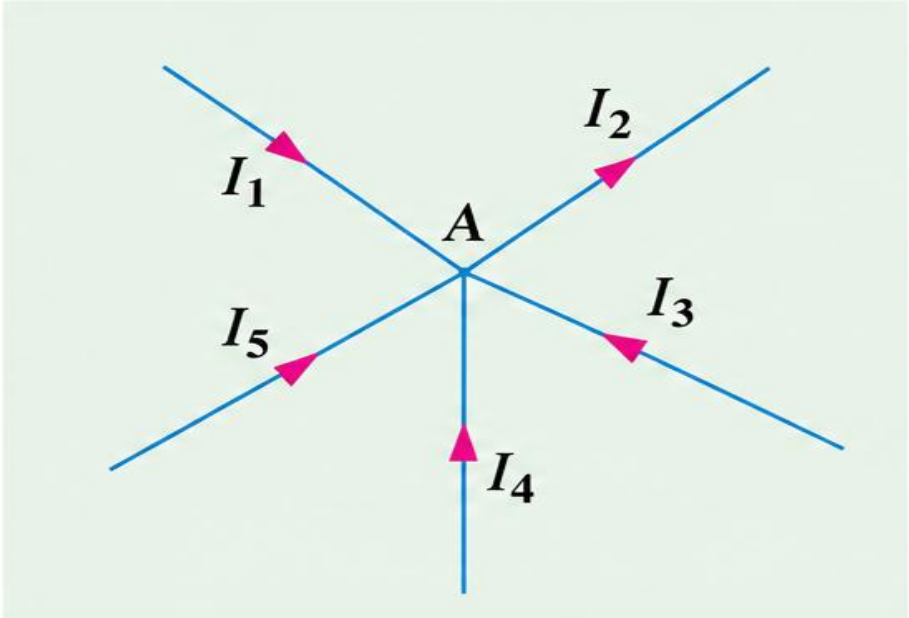


Kirchhoff's Laws
Kirchhoff's Current Law or Point Law (KCL)

Statement:
"The algebraic sum of all currents entering and leaving a node must equal to zero"

In other words, the total current leaving a junction is equal to the total current entering that junction.

$$I_1 - I_2 - I_3 + I_4 - I_5 = 0$$



Kirchhoff's Laws
Kirchhoff's Voltage Law or Mesh Law (KVL)

Statement:
"The algebraic sum of all voltage drops around any closed loop is zero"

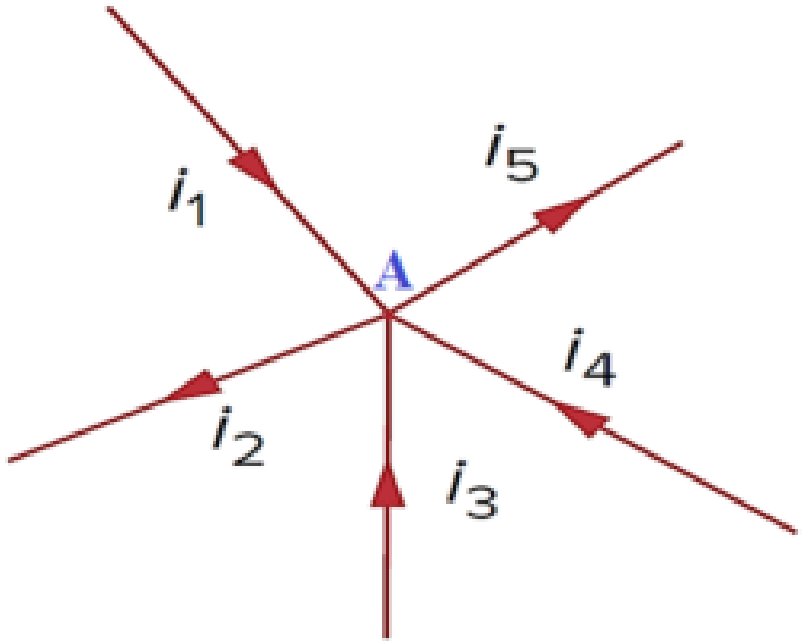
In other words,

$\Sigma\, IR + \Sigma\, e.m.f. = 0$ ...round a mesh

KIRCHHOFF'S CURRENT LAW (KCL)

- **Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.**

- Fig. below shows that currents are entering and leaving the **Node A**.

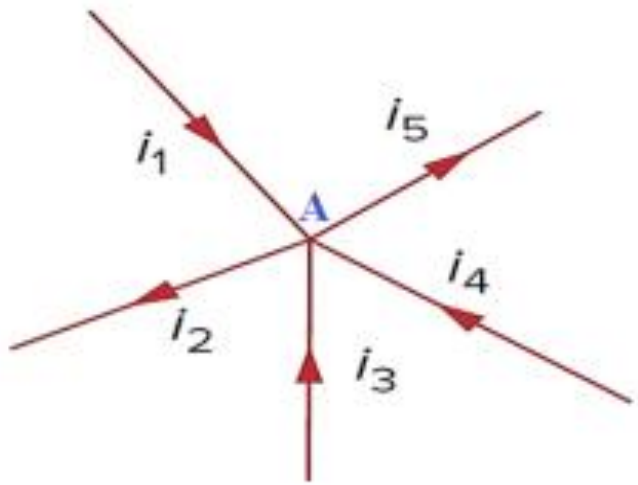


KIRCHHOFF'S CURRENT LAW (KCL)

- Before applying **KCL** sign convention is very important**.** The following sign convention is used:

Current Entering node "-ve"
Current Leaving node "+ve"

- Applying **Kirchhoff's current law** with sign convention one has,



$$-I_1 + I_2 - I_3 - I_4 + I_5 = 0$$

KIRCHHOFF'S CURRENT LAW (KCL)

Kirchhoff's Current Law (KCL) is based on the principle of conservation of charge.

It states that the algebraic sum of all currents at a node is zero.

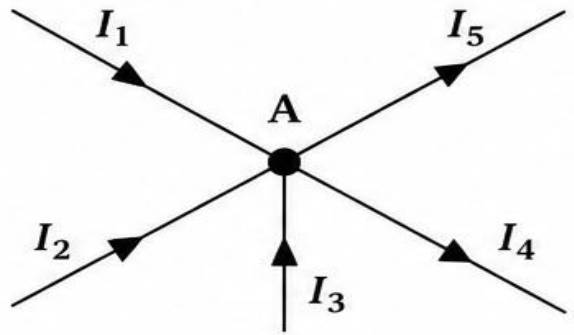
SIGN CONVENTION USED IN KCL

Current Entering node	=	**"-ve"**
Current Leaving node	=	**"+ve"**

APPLYING KCL WITH SIGN CONVENTION

Consider a node A with five currents I_1, I_2, I_3, I_4 and I_5 as shown below.

Currents entering the node (I_1, I_2, I_3) are taken as –ve.



Currents leaving the node (I_4, I_5) are taken as +ve.

Applying Kirchhoff's current law,

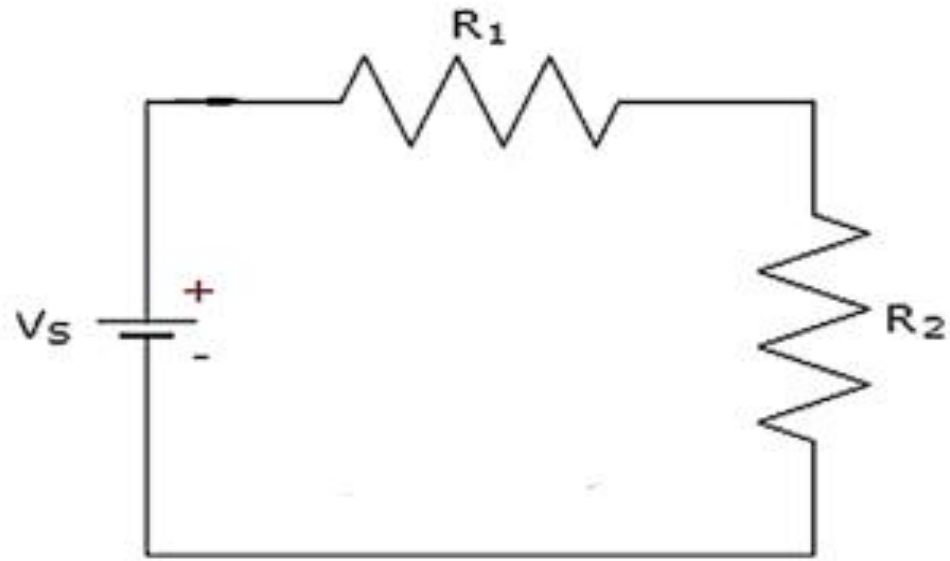
$$-I_1 - I_2 - I_3 + I_4 + I_5 = 0$$

Alternatively, Kirchhoff's current law is also stated as,

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

KIRCHHOFF'S VOLTAGE LAW (KVL)

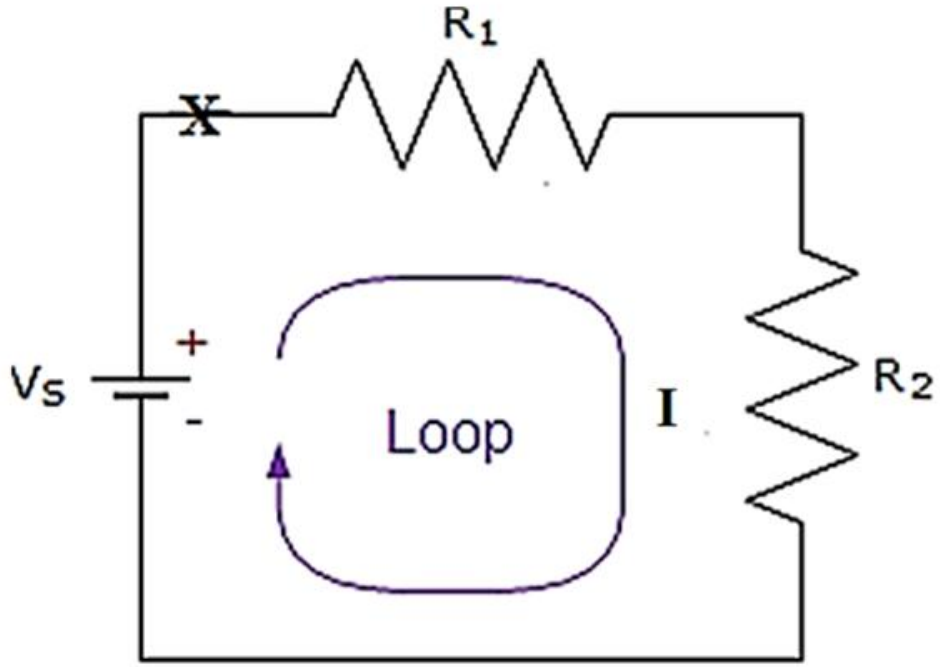
- **Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.**
- To illustrate KVL, consider the circuit.



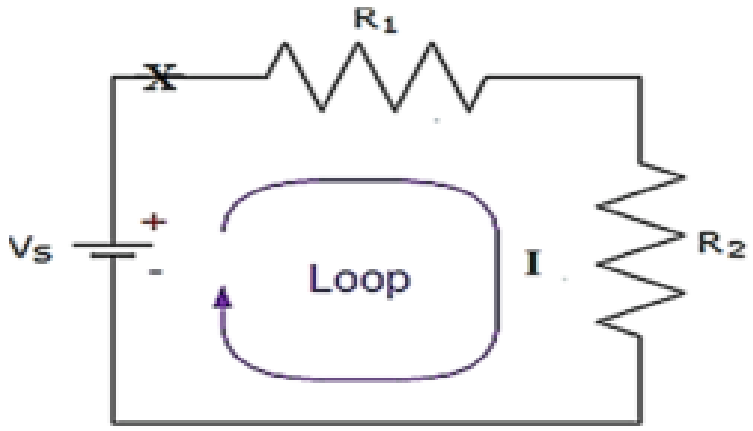
KVL

Step 1: Let us assume clockwise current in the loop.

Step 2: Moving in direction of current and writing KVL equation. Let us assume we start moving from location **X** towards right.



KVL



$$V_S = IR_1 + IR_2$$

If voltage drop across resistances are termed as V_1 and V_2 ,

$$V_S = V_1 + V_2$$

Furthermore,

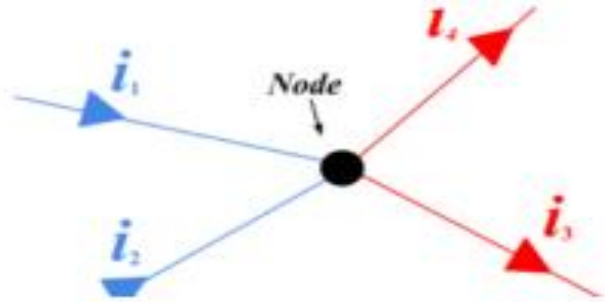
$$V_S - V_1 - V_2 = 0$$

shows total voltage is sum of drops across each resistances which is known as **Kirchhoff's Voltage Law.**

KCL AND KVL

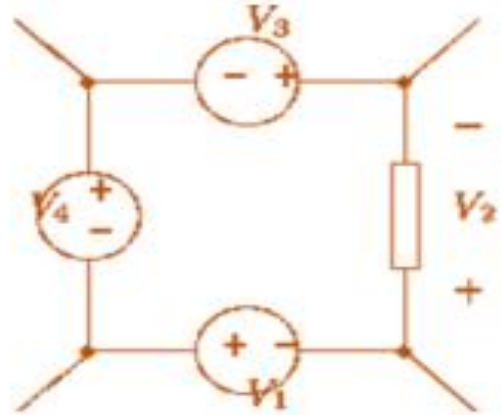
- Mathematically KCL is written as,

$$\sum_{n=1}^{N} I_n = 0$$



where, N is number of branches, I_n is nth branch current entering/leaving the node. Similarly KVL for voltage across total M elements,

$$\sum_{m=1}^{M} V_n = 0$$

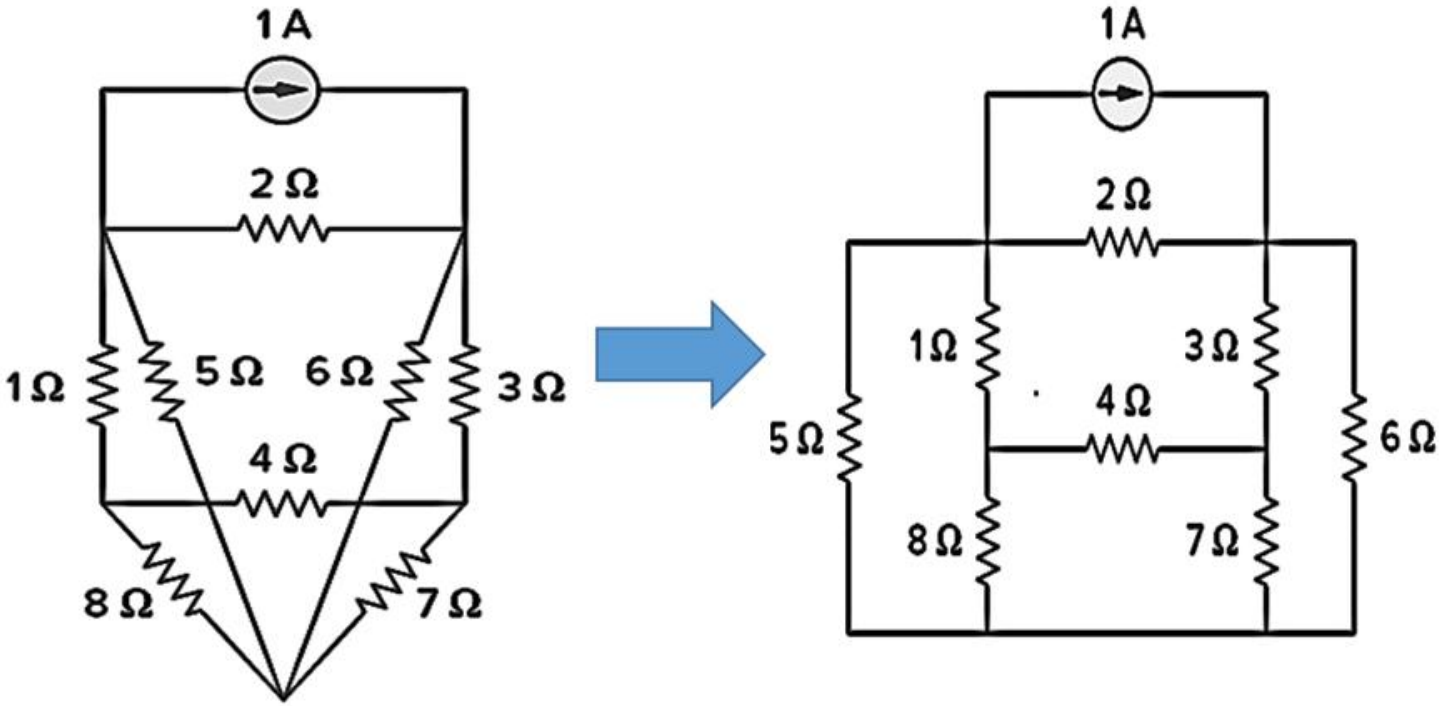


MESH ANALYSIS

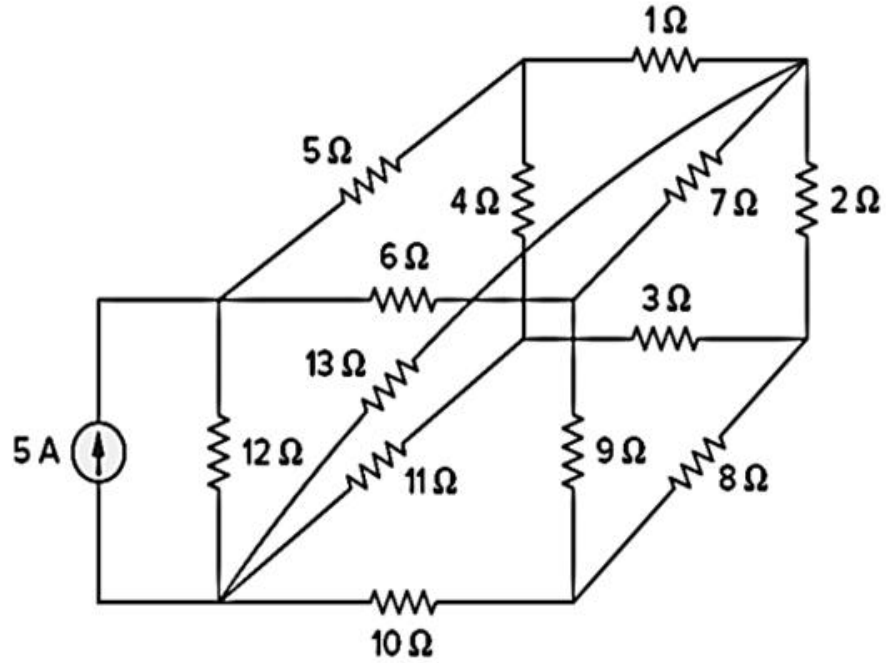
- **Loop:** It is a closed path with no node passed more than once.
- **Mesh**: A mesh is a loop that does not contain any other loop within it.
- **Mesh analysis** is only applicable to a **circuit** that is **planar**.
- **Planar Circuit**: A planar circuit is one that can be drawn in a plane with no branches crossing one another.
- **Non-Planar Circuit**: A nonplanar circuit is one that can be drawn in a plane with branches crossing one another.

MESH ANALYSIS

- Some circuits may have crossing branches, but they can be redrawn as planar.



PLANAR CIRCUIT



NONPLANAR CIRCUIT

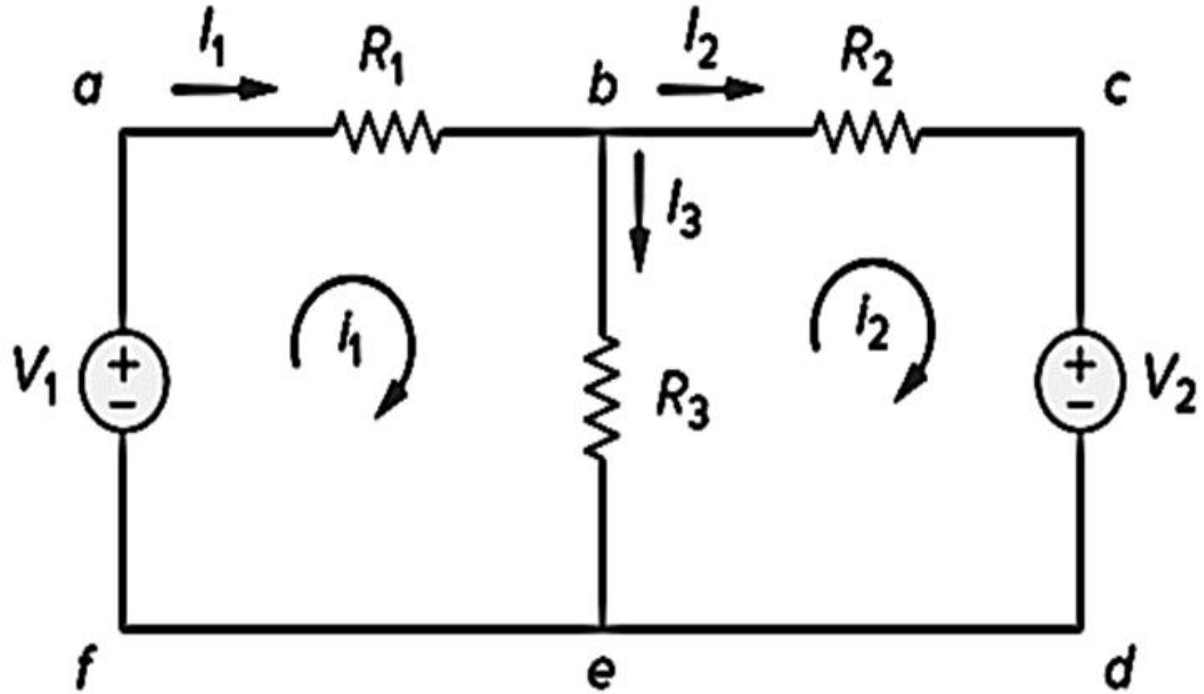
MESH ANALYSIS

- Let us learn mesh analysis to planar circuits that do not contain current sources.
- In the mesh analysis of a circuit with n meshes, we take the following three steps:

- Assign mesh currents i_1, i_2, . . . , in to the ***n*** meshes.
- Apply **KVL** to each of the ***n*** meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
- Solve the resulting ***n*** simultaneous equations to get the mesh currents.

MESH ANALYSIS

- Consider circuit shown below with two meshes, currents are assigned in clockwise direction in mesh. Let us analyze mesh for calculating current using KVL in each mesh.



MESH ANALYSIS

- Applying KVL in mesh 1.

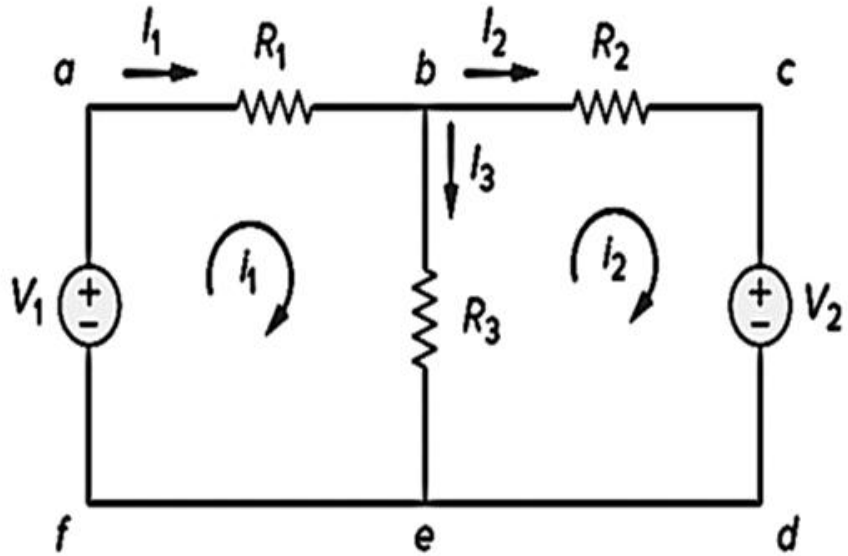
$$-V_1 + R_1 i_1 + R_3 \, i_3 = 0 \qquad (1)$$

- Applying KCL at node b,

$$-i_1 + i_2 + i_3 = 0 \Rightarrow i_3 = i_1 - i_2 \qquad (2)$$

- Substituting (2) in (1) ,

$$\boxed{V_1 = \left(R_1 + R_3\right) i_1 - R_3 \, i_2} \qquad (I)$$



MESH ANALYSIS

- Applying KVL in mesh 2.

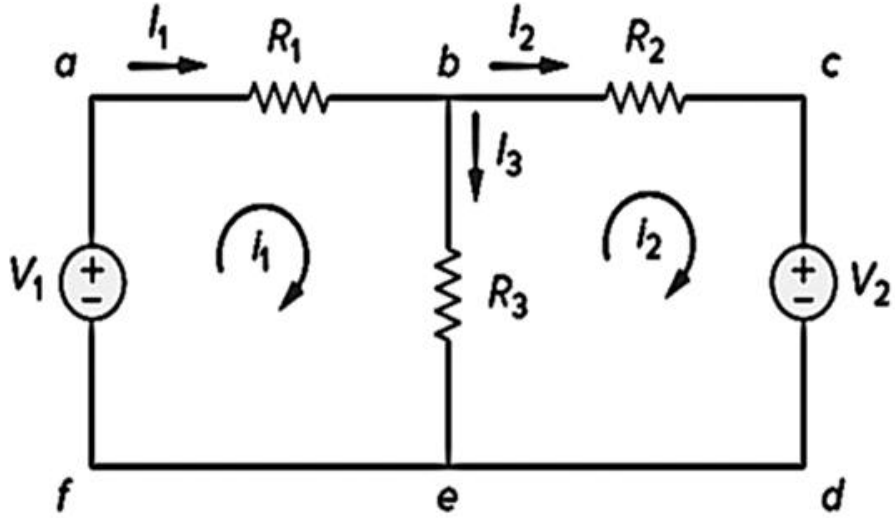
$$R_2 i_2 + V_2 - R_3 i_3 = 0 \qquad (1)$$

- Replacing i_3,

$$R_2 i_2 + V_2 - R_3 (i_1 - i_2) = 0 \qquad (2)$$

- Substituting (2) in (1) ,

$$\boxed{V_2 = -R_3 i_1 + (R_1 + R_2) i_2} \qquad (II)$$



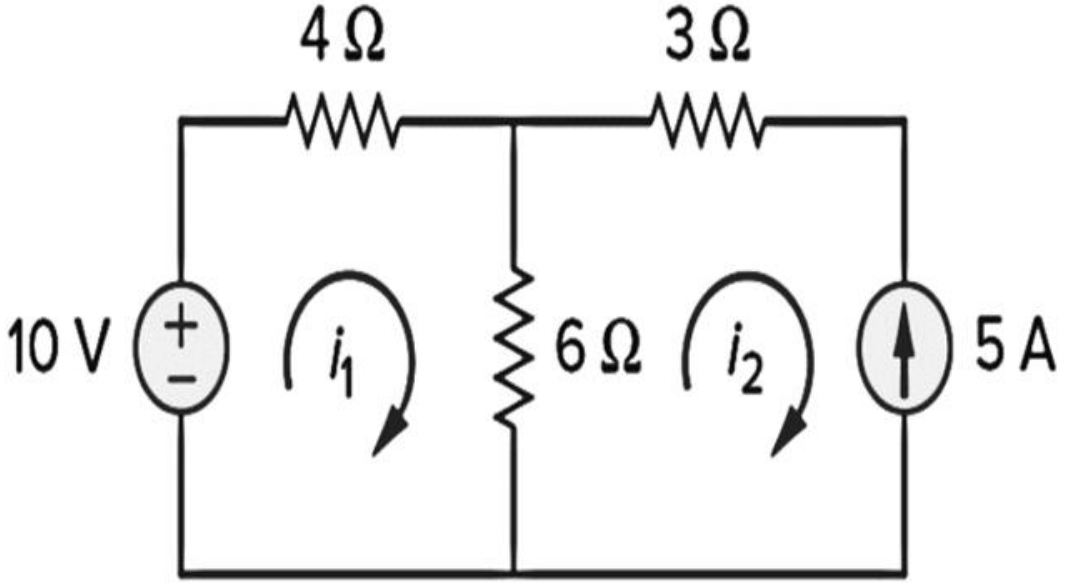
$$V_2 = I_1 R_3 - I_2 (R_2 + R_3)$$

MESH ANALYSIS

- Eqs. (I) and (II) are mesh 1 and 2 equations.
- These can be solved to obtain two unknowns i_1 *and* i_2.
- Moreover, i_3 can also be obtained which is difference of two.
- Hence, two mesh circuit currents are calculated when voltage sources and resistance values are given using steps of mesh analysis.
- In case of three meshes in a circuit there are three equations and we can use Cramer's rule to solve 3 linear equations.

MESH ANALYSIS

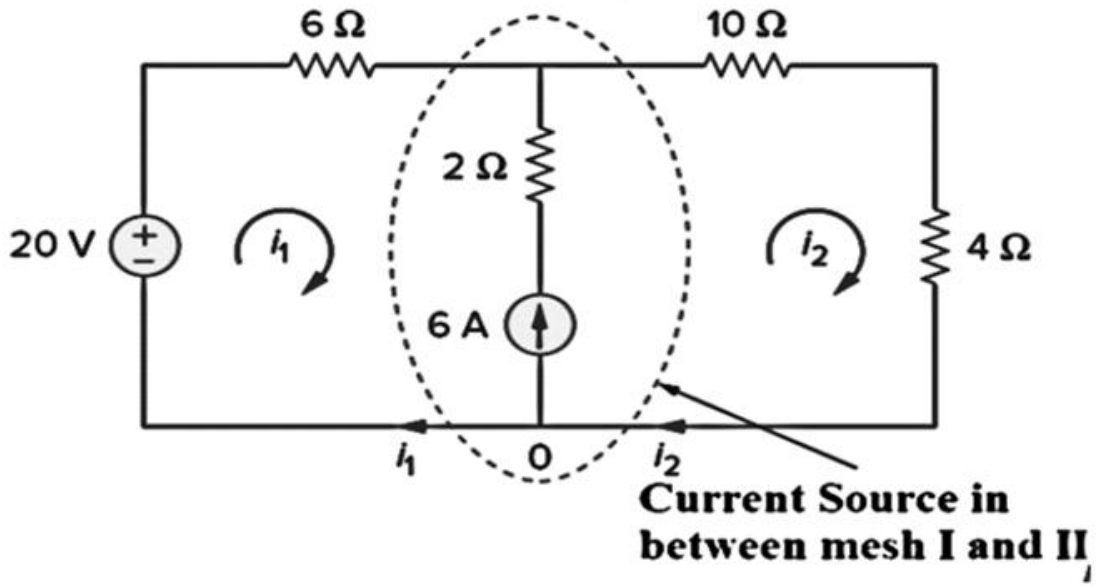
- **Case 1** makes calculation easier as current source itself give value of current flowing in that mesh.



- Hence, i_2 becomes -5 A , putting value of i_2 in KVL, current of mesh I is easily calculated.

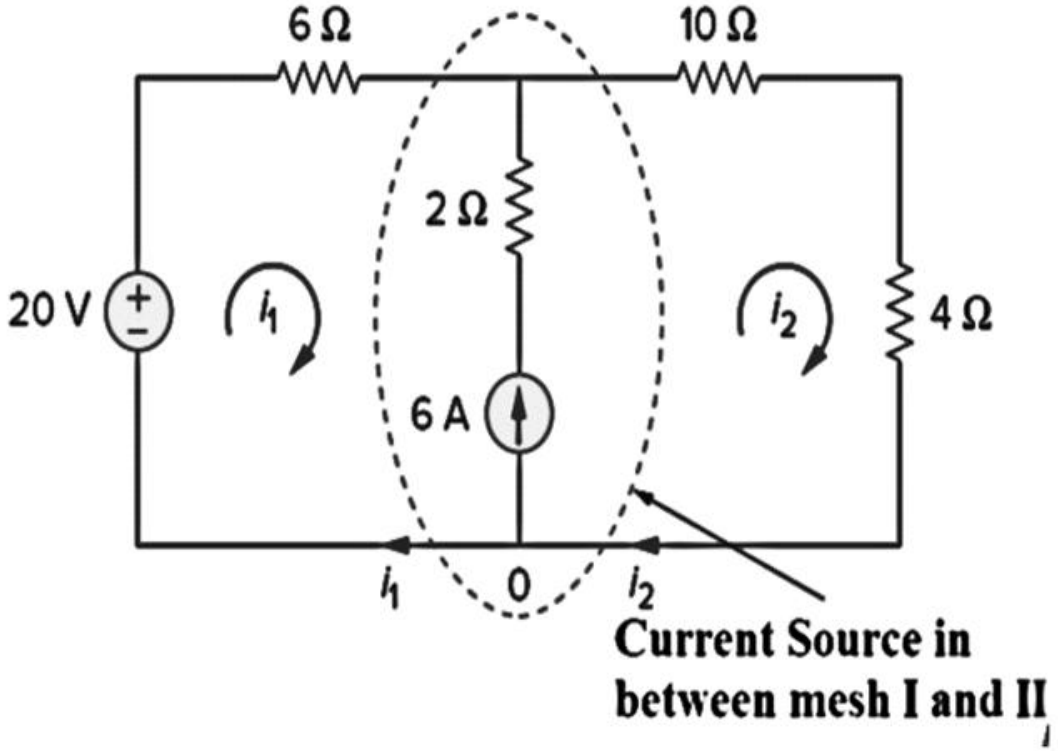
SUPER MESH CONCEPT

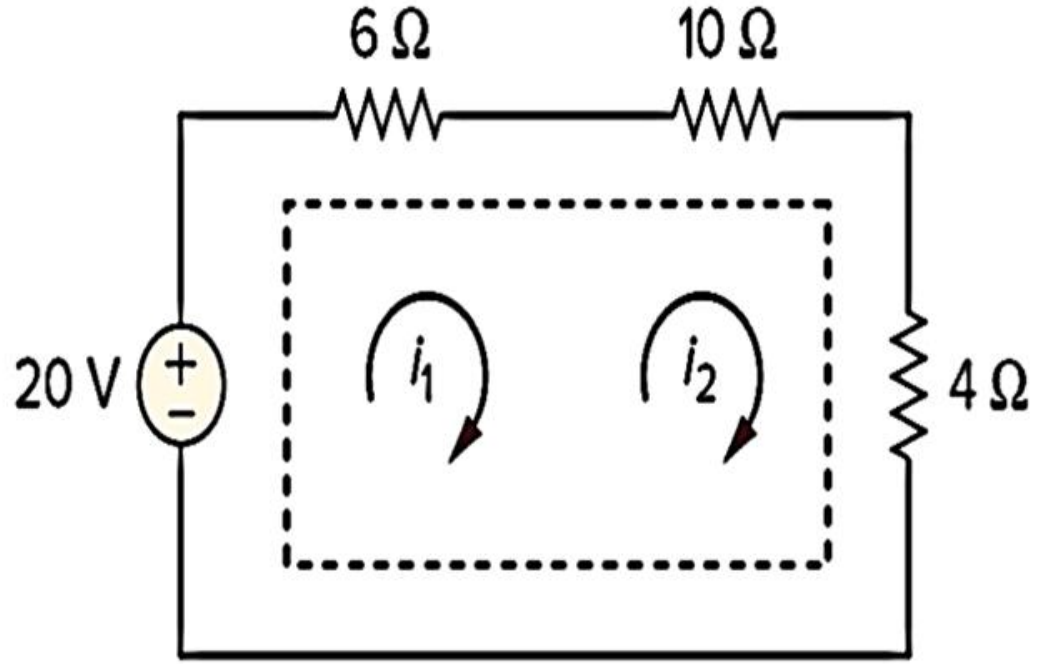
- When current source (dependent/independent) occurs between two meshes we apply super mesh concept to solve the circuit.
- We create a **super mesh** by **excluding the current source and any elements connected in series with it** while taking KVL.
- Let us solve such circuit shown below.



SUPER MESH CONCEPT

- Converting Fig. having two meshes into a circuit having single super mesh.



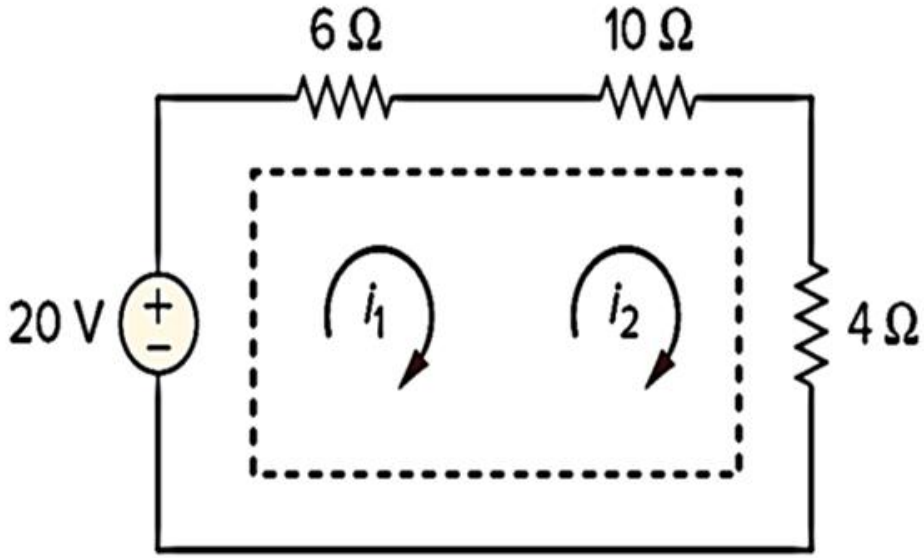


SUPER MESH CONCEPT

- Applying KVL to the supermesh.

$$-20+6i_1 +10i_2+4i_2=0$$

$$\Rightarrow \boxed{6i_1 +14i_2=20}$$



- Applying KCL at bottom node,

$$\boxed{i_2=i_1+6}$$

- We can easily solve above two equations to obtain i_1 *and* i_2.
- A supermesh requires the application of both KVL and KCL.

Q] Three branches meet at a junction in an LED circuit. A current of **2 A** enters the junction. Currents of **0.8 A** and **0.7 A** leave through two branches.

Question: Find the current leaving through the third branch.

Using KCL:

$I_{in}=I_1+I_2+I_3$ $2=0.8+0.7+I_3$ **Answer:**

$I_3=0.5A$

Q] A **12 V battery** is connected to an LED circuit. The voltage drop across the resistor is **7 V**.

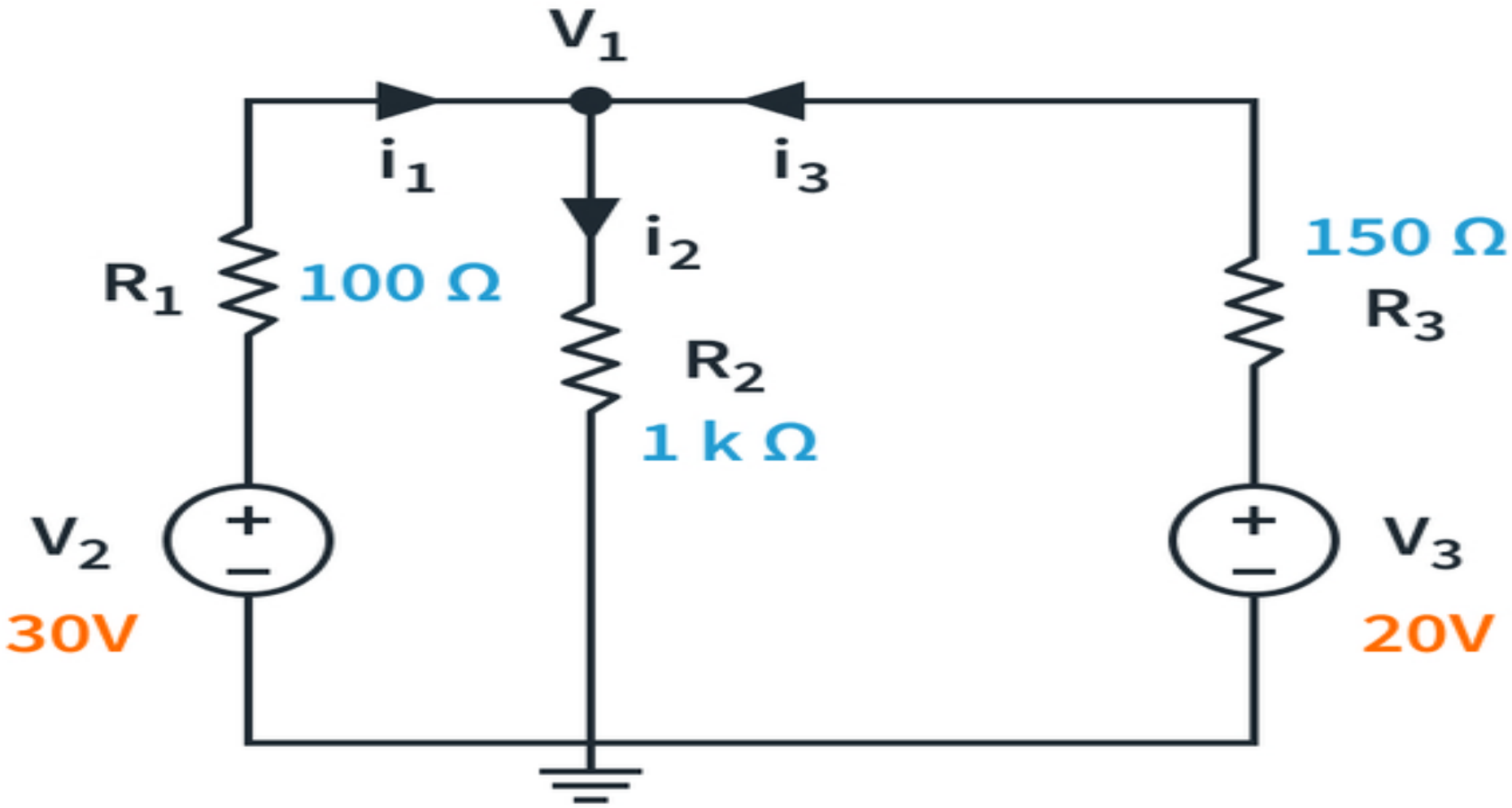
Question: Find the voltage drop across the LED.

Using KVL:

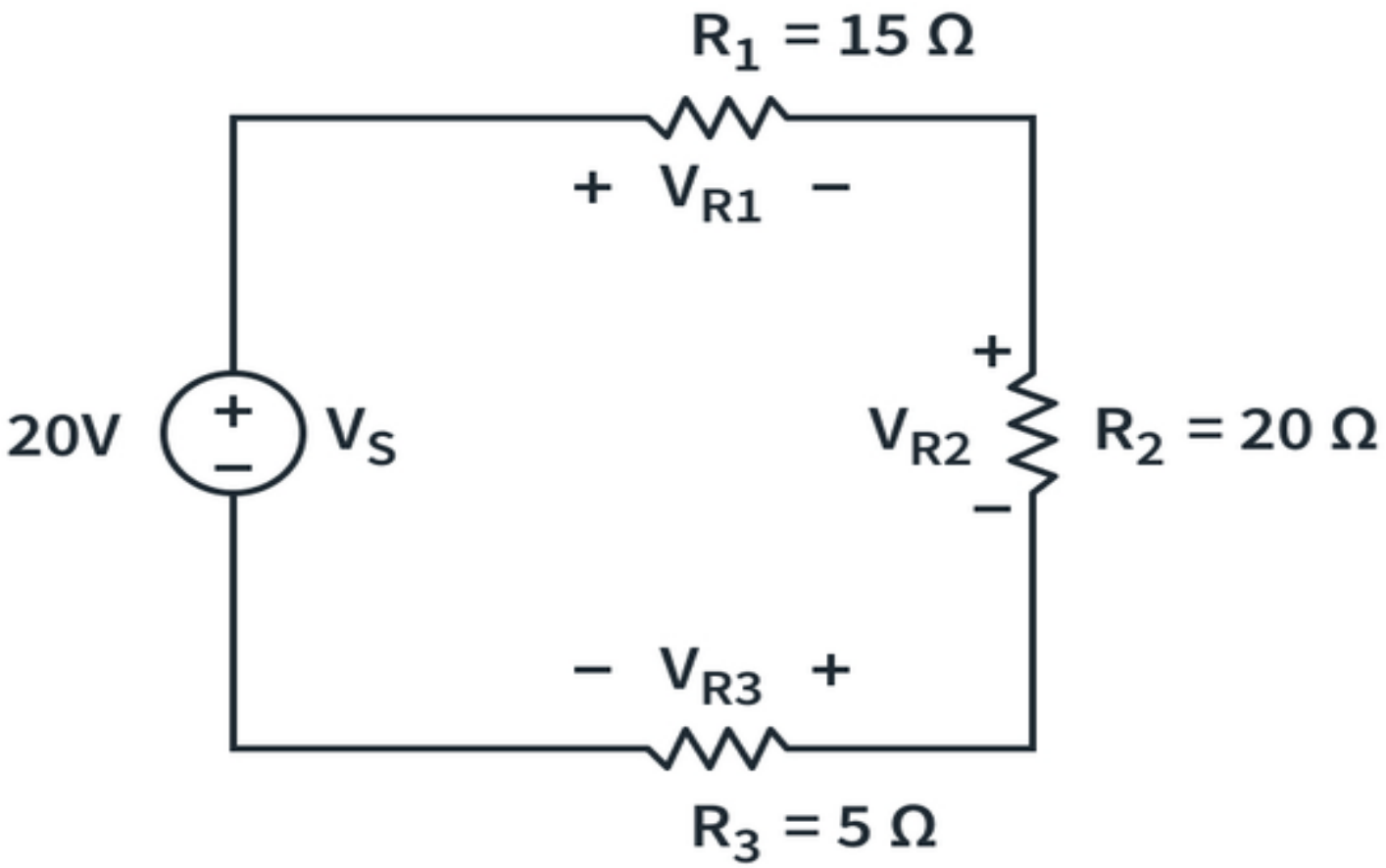
$12-7-V_{LED}=0$ **Answer:**

$V_{LED}=5V$

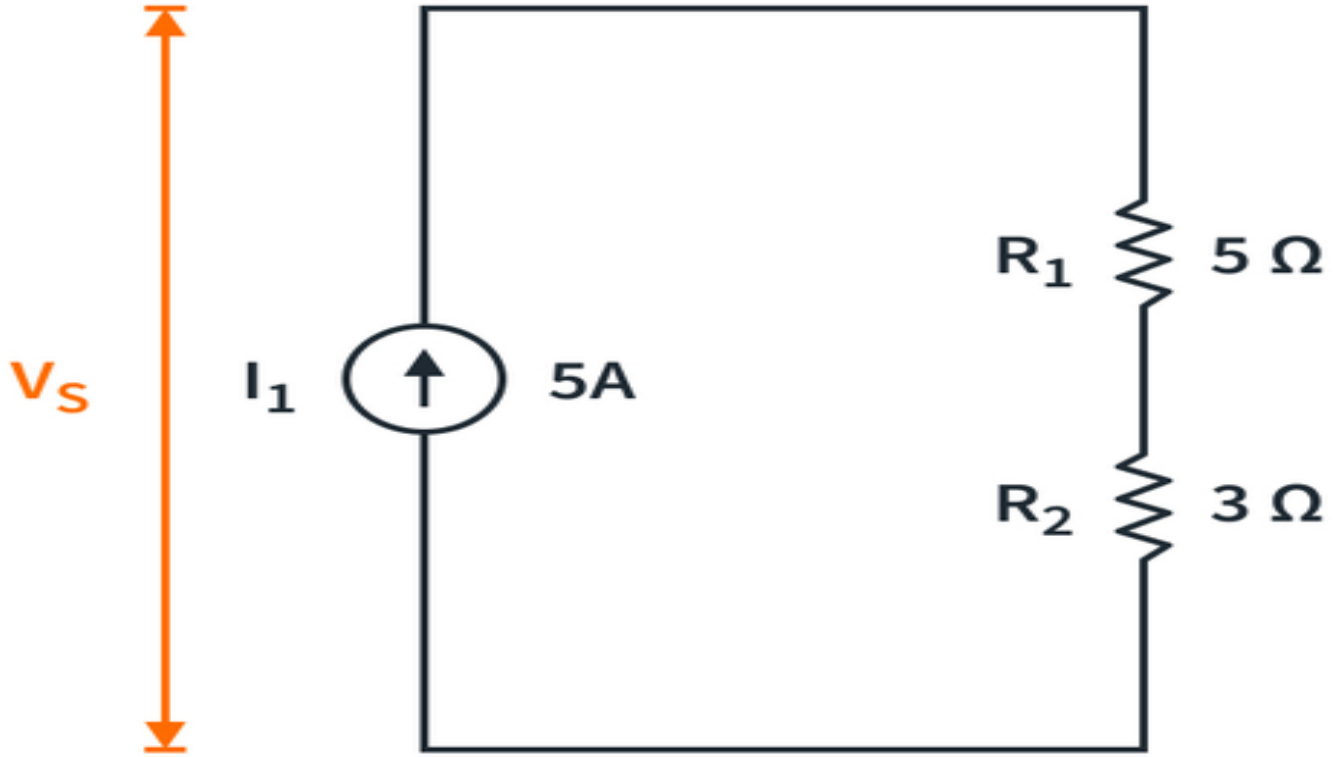
Problem 1: Calculate Branch Currents I1,I2,I3



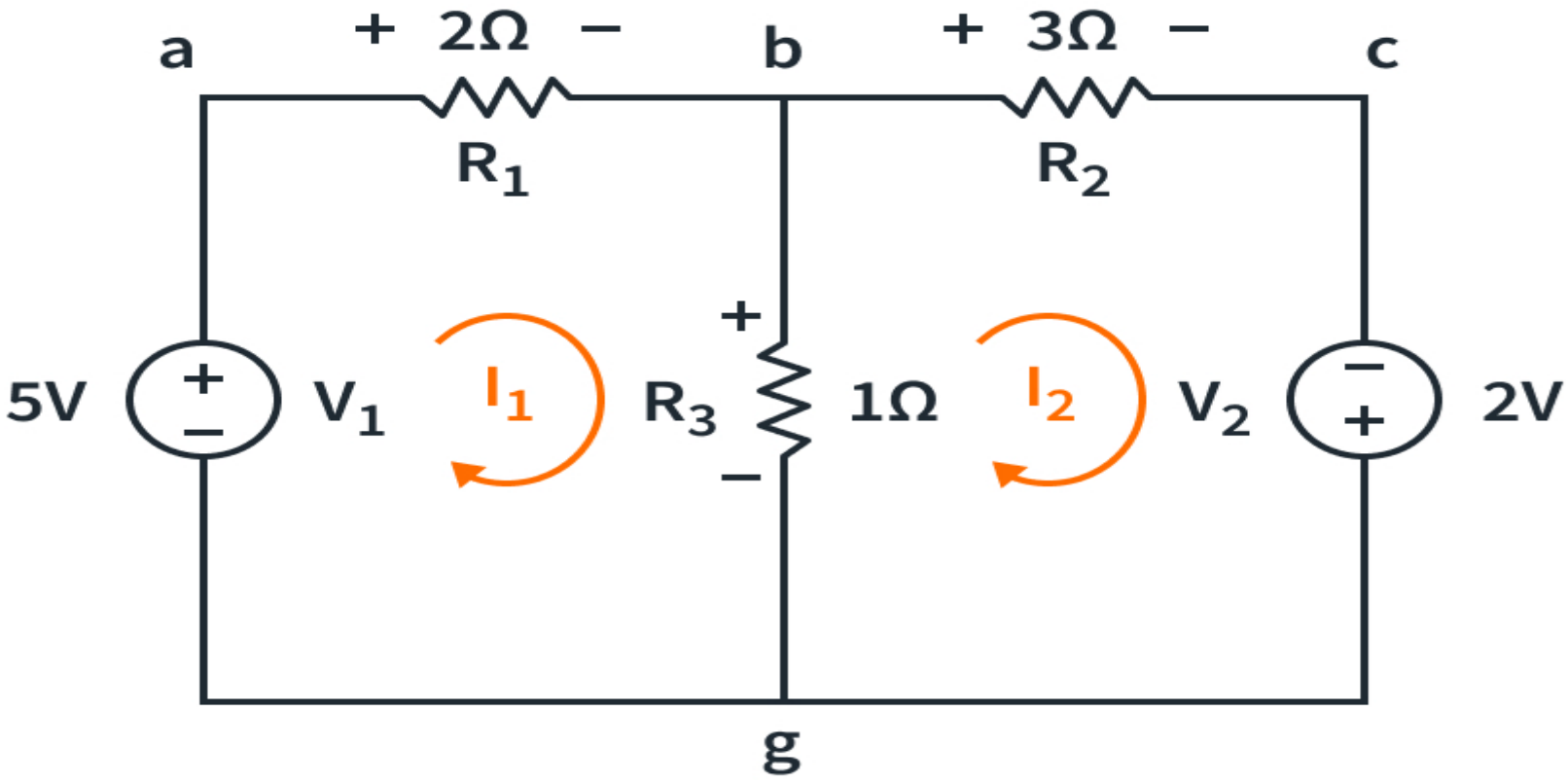
Problem 2. Prove that the KVL is validated



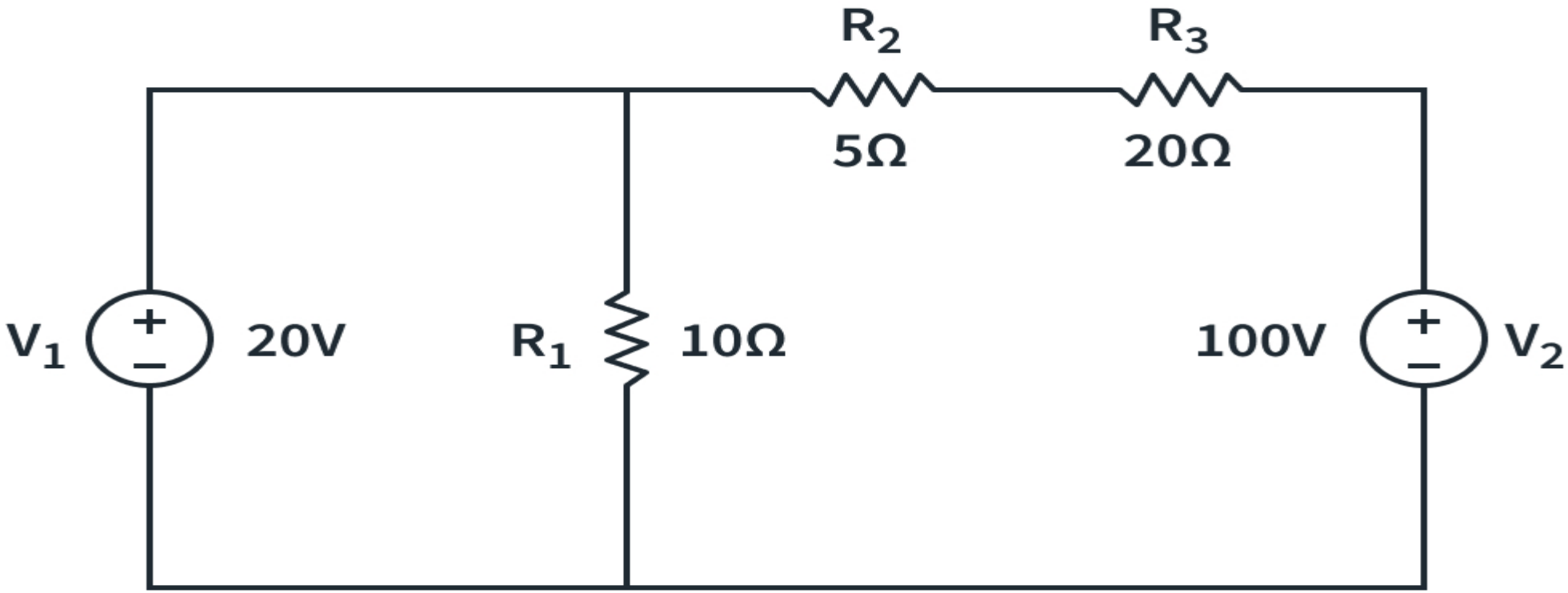
Consider the following circuit with resistors R1 = 5 ohms and R2 = 3 ohms and a current source I1 = 5A. Determine the voltage drop across the current source using KVL.



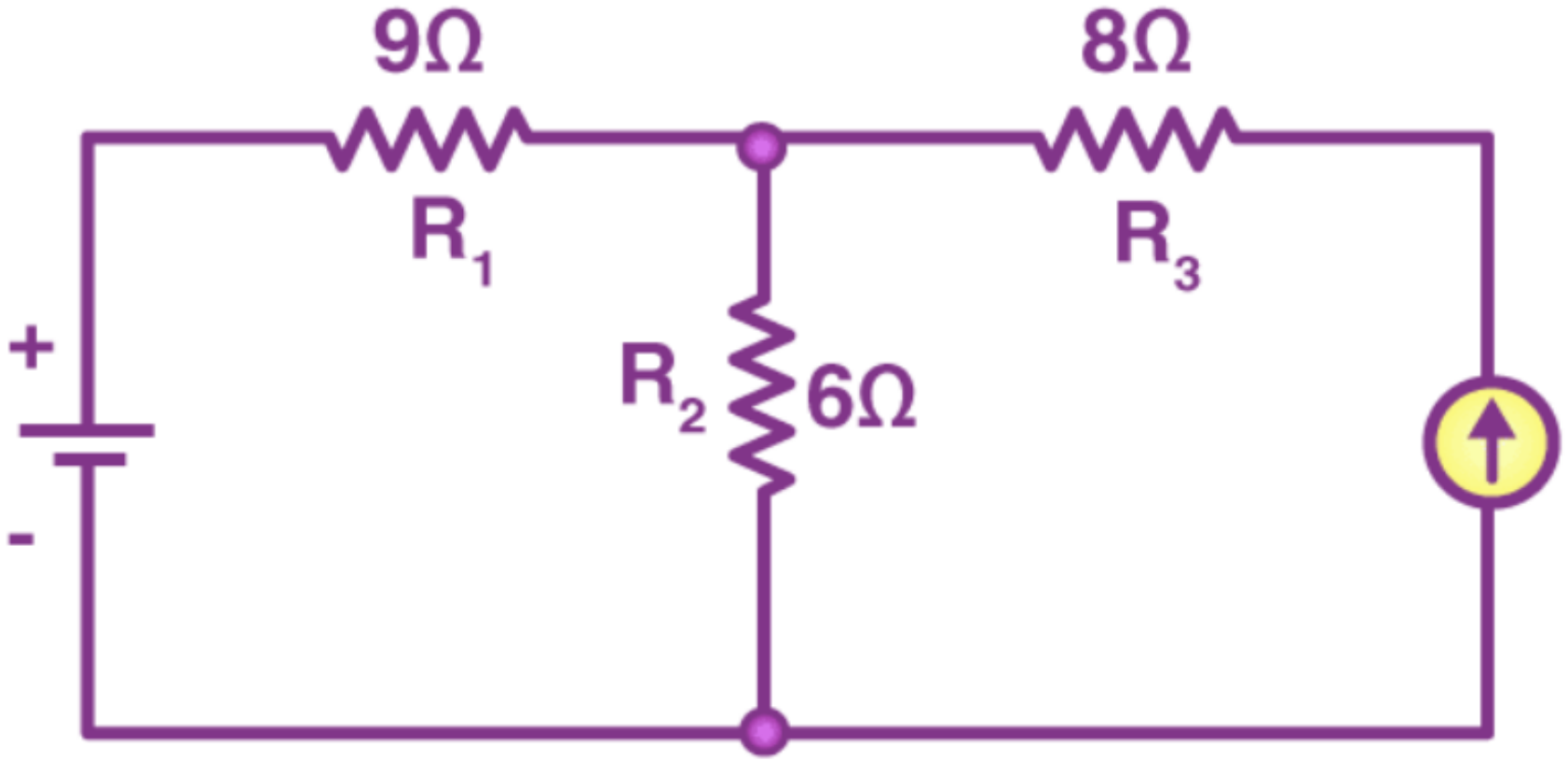
Consider the following circuit with two loops. There are two voltage sources V1 = 5V and V2 = 2V. Currents I1 and I2 pass through loops 1 and 2 respectively with resistances R1 = 2 ohms, R2 = 3 ohms and R3 = 1 ohms. Using KVL, find currents I1 and I2.



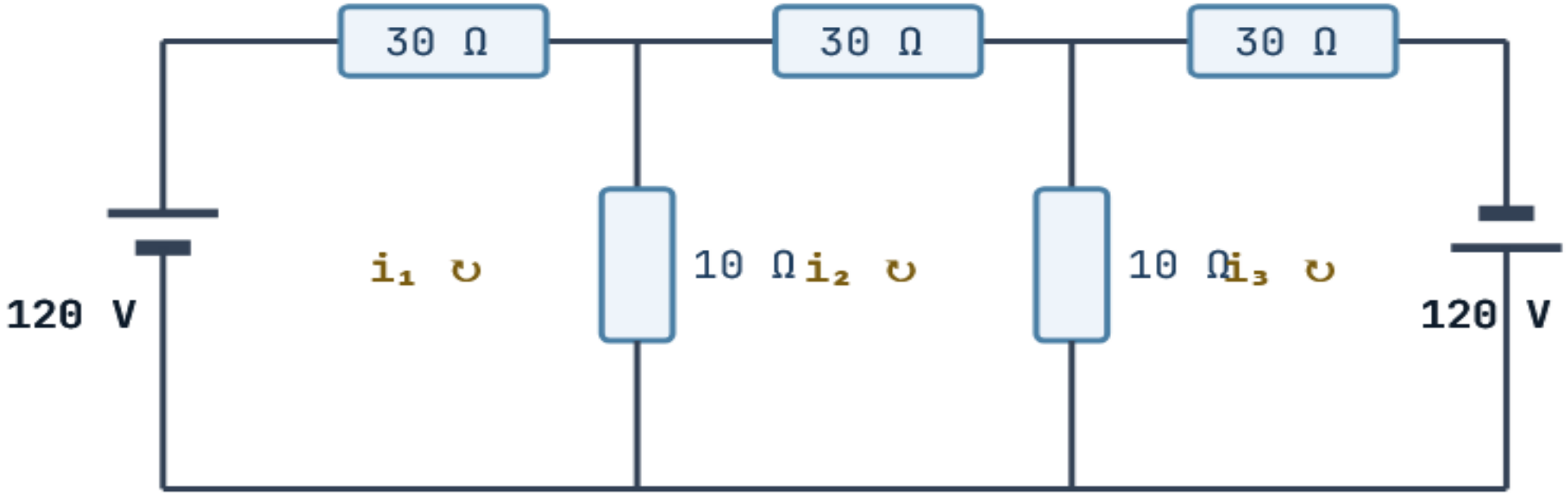
Consider the following circuit. Use KCL and KVL to find the currents through each branch of the circuit as well as the voltages across each element.



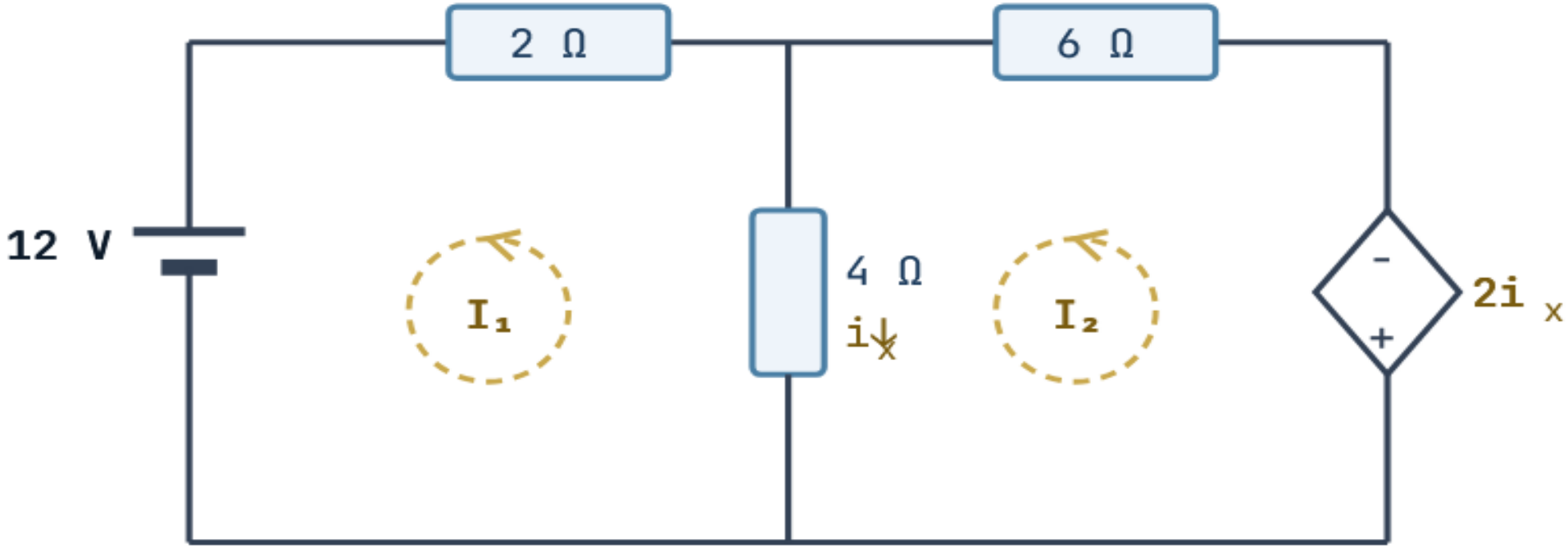
In the given circuit 90v is the battery value, 5A is the current source and the three resistors are 9 ohms, 6 ohms, and 8 ohms. Using mesh analysis, determine the current across each resistor and potential difference.



A three-mesh network has a 120 V source in mesh 1 and another 120 V source in mesh 3, with mesh 2 sitting between them. The mesh resistances total 40 Ω, 50 Ω and 40 Ω respectively, and meshes 1–2 and 2–3 each share a 10 Ω resistor. Find the three mesh currents.

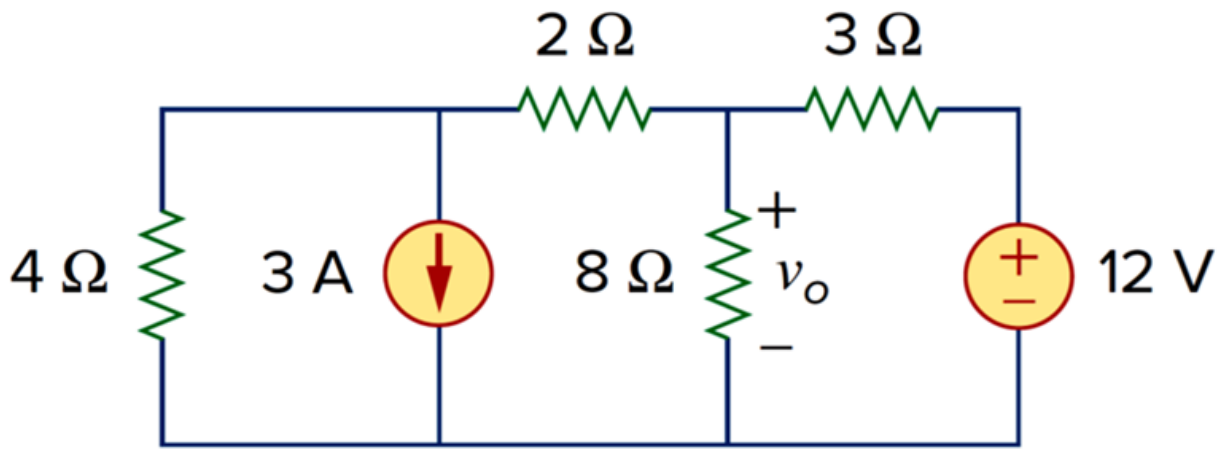


A 12 V source drives mesh 1 through a 2 Ω resistor. The two meshes share a 4 Ω resistor carrying downwards. Mesh 2 contains a 6 Ω resistor and a current-controlled voltage source of value , oriented as a rise in the clockwise direction. Find both mesh currents and the power in the dependent source.



EXERCISE / NUMERICAL ANALYSIS

Q Use source transformation to find v_o in the circuit of below figure.

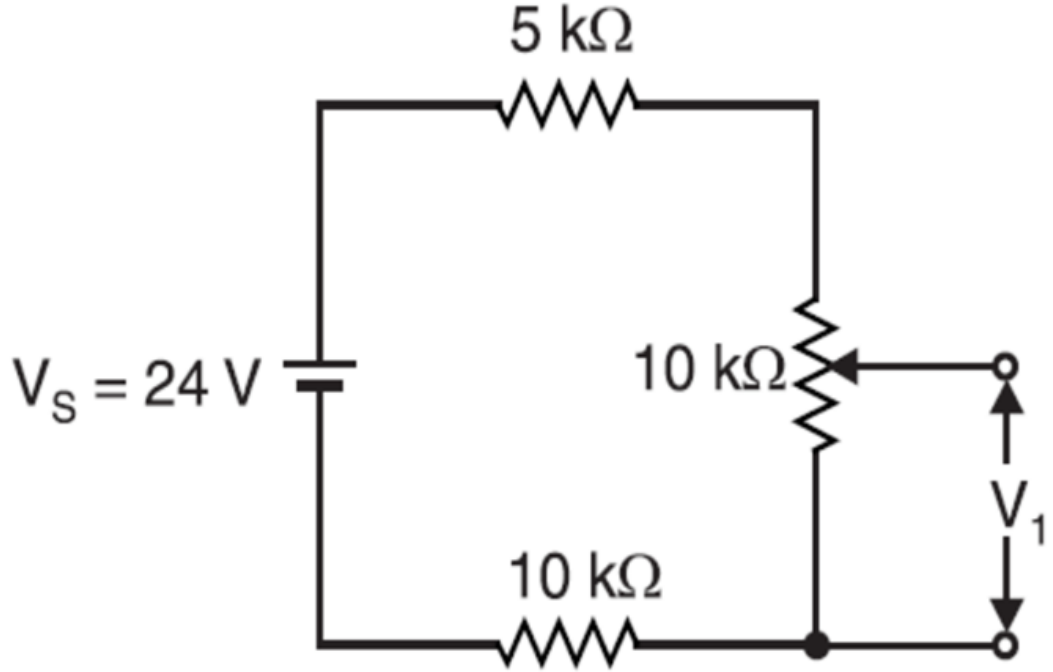




EXERCISE/NUMERICAL ANALYSIS



Q. A 10 kΩ potentiometer connected in a series circuit as an adjustable voltage divider. What total range of voltage V1 can be obtained by adjusting the potentiometer through its entire range.

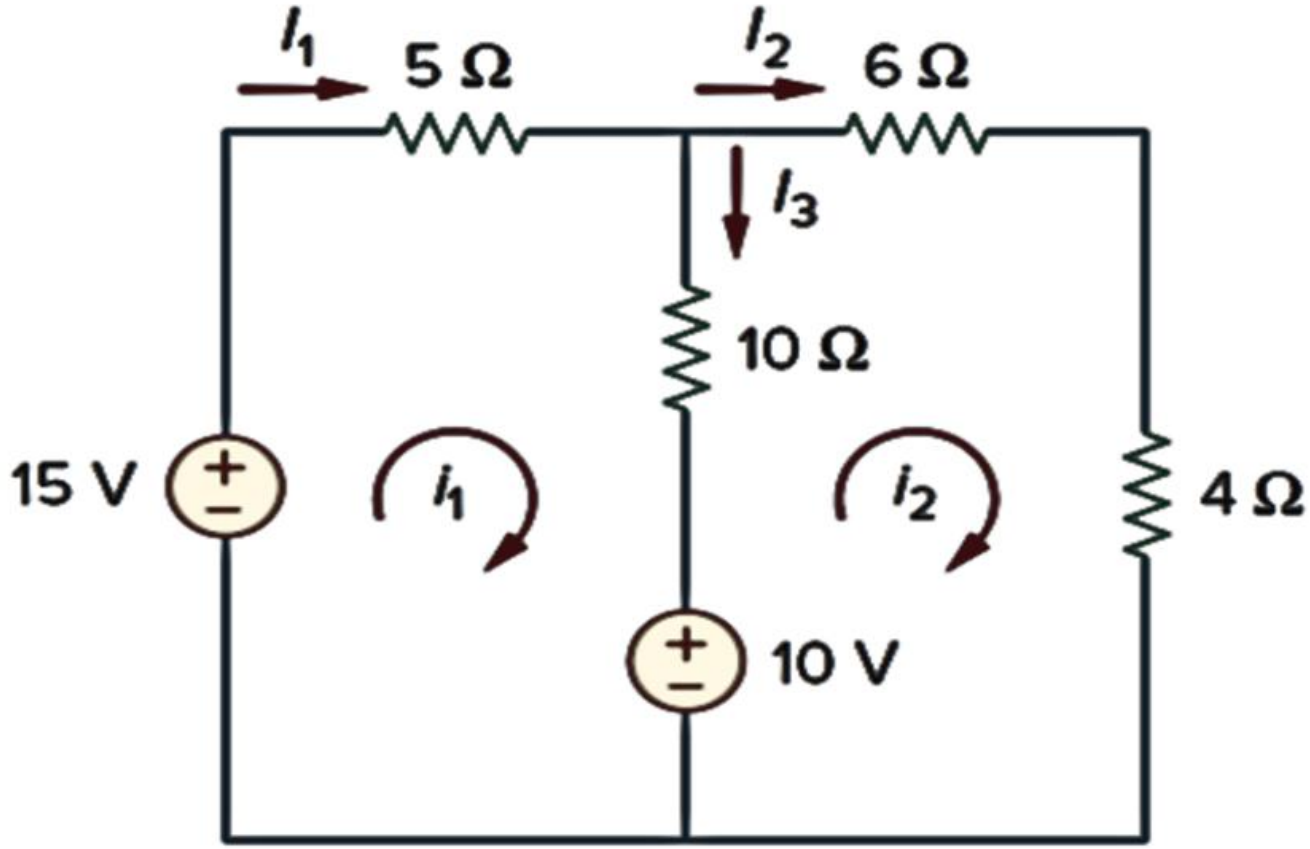




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1, i_2, i_3 using mesh analysis.

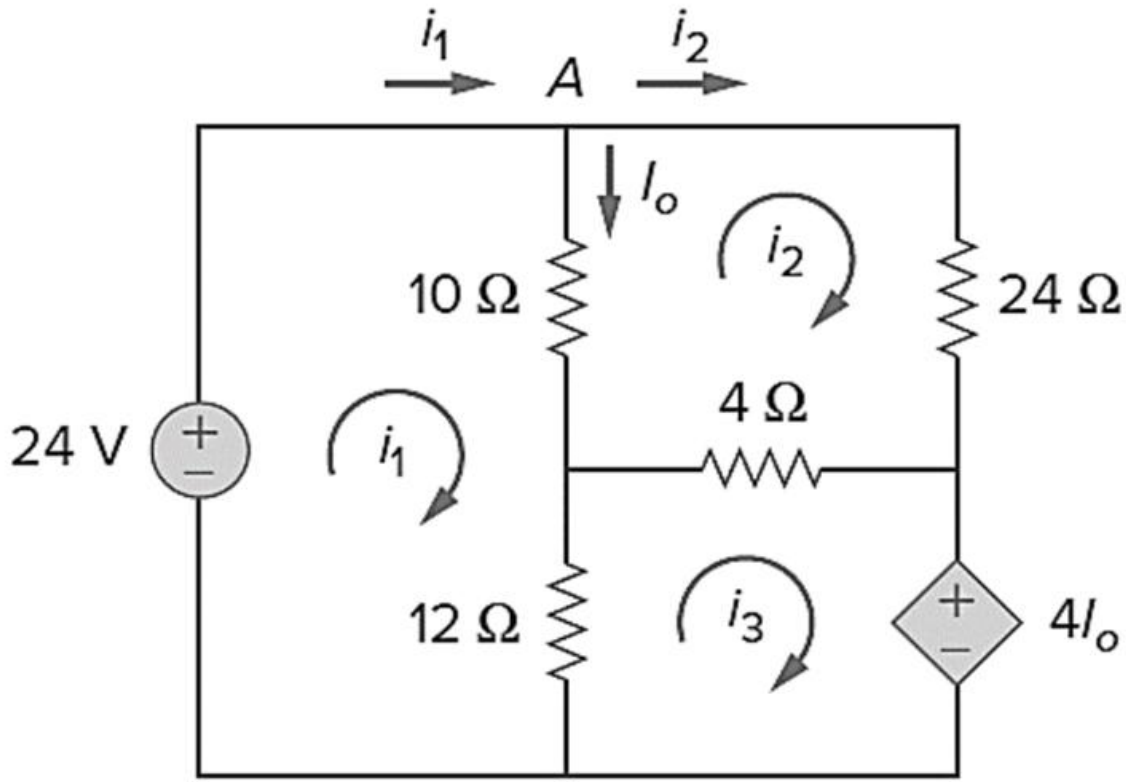




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1, i_2, i_3 *and* i_o using mesh analysis.

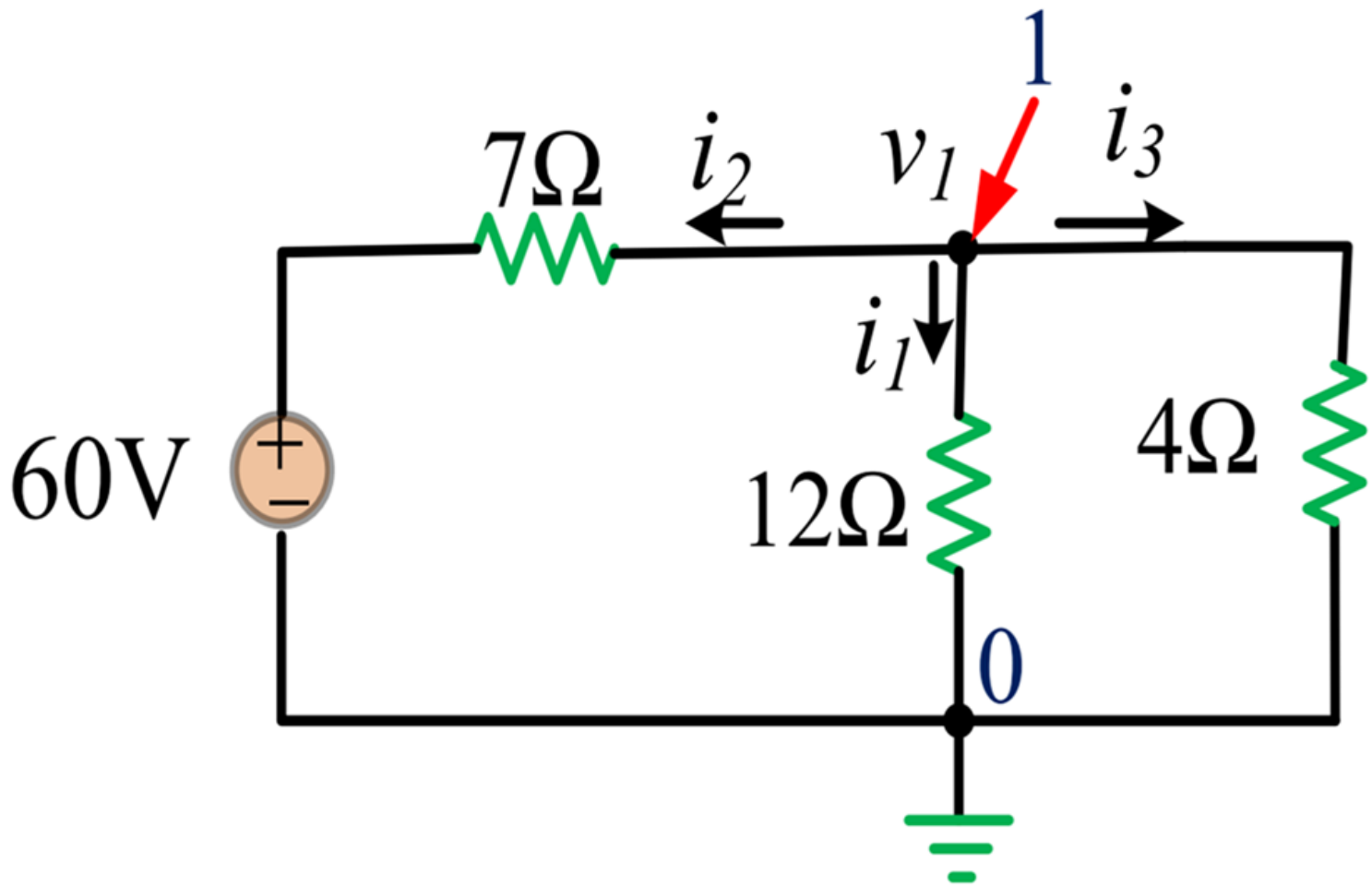




EXERCISE/NUMERICAL ANALYSIS



Q Calculate Node Voltages and Branch Currents in the given circuit.



Q] Challenging Problem: Consider a circuit with a 30 V source. The source current reaches node A and divides into two branches. Branch 1:R1=5Ω
Branch 2:R2=10Ω,R3=5Ω
The two branches reconnect at node B.

Questions:

1. Draw the circuit.
2. Determine the equivalent resistance of Branch
3. late the current in each branch using Ohm's Law.
4. Apply KCL at node A.
5. Calculate the voltage drop across every resistor.
6. Apply KVL around Branch 1.
7. Apply KVL around Branch 2.